

# TRANSFORMER OIL ANALYSIS

Fikri Khoirur Rizal

PT TRANSFORMATOR SERVIS INDONESIA

# TRANSFORMER OIL ANALYSIS

Pengujian minyak pada trafo bertujuan untuk mengetahui kondisi trafo, kondisi isolator kertas dan kualitas minyak trafo sendiri. Dari hasil pengujian minyak trafo dapat dilakukan tindakan lanjutan untuk menjaga kehandalan dari trafo dan memperpanjang lifetime trafo.

# REASON & BENEFIT

- ▶ Menjaga kehandalan trafo
- ▶ Memperpanjang masa pakai trafo
- ▶ Menjadi acuan rencana dan skala prioritas dalam maintenance

# FUNGSI MINYAK TRAF0



- ▶ Melindungi isolator kertas
- ▶ Isolator elektrik
- ▶ Penghantar panas
- ▶ Indikator kondisi trafo

# TRANSFORMER OIL ANALYSIS

|                               |                                                                                                                                                                                                                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kondisi Transformator         | Dissolved Gas Analysis                                                                                                                                                                                                                                                                                      |
| Kondisi Isolator Kertas       | Furan Analysis                                                                                                                                                                                                                                                                                              |
| Kualitas Minyak Transformator | <div>Oil Quality Analysis<ul style="list-style-type: none"><li>- Breakdown Voltage</li><li>- Water Content</li><li>- Acidity</li><li>- Color &amp; Appearance</li><li>- Interfacial Tension</li><li>- Dielectric Dissipation Factor</li><li>- Corrosive Sulphur</li><li>- Inhibitor Content</li></ul></div> |

# DISSOLVED GAS ANALYSIS IEEE C57.104-2019

Pengujian untuk mengetahui kondisi  
transformator

DGA is one of the most widely used diagnostic tools for transformer condition assessment because experience has proven it to be an effective tools.

IEEE C57.104 - 2019

# Perubahan isi Standar

- ▶ Tabel 4 kondisi, diganti dengan table 1 dan table 2.
- ▶ Dihapuskan interpretasi TCG dan TDCG.
- ▶ Penambahan rate kenaikan gas pada table 3 dan table 4.
- ▶ Flowchart baru untuk interpretasi.
- ▶ Penambahan jenis gangguan.
- ▶ Duval Triangle 1, 4 dan 5 dan Duval Pentagon 1 dan 2.
- ▶ Interpretasi CO & CO<sub>2</sub>.

**Table 1.—Dissolved Key Gas Concentration Limits in Parts Per Million (ppm) [4]**

| Status      | Hydrogen (H <sub>2</sub> ) | Methane (CH <sub>4</sub> ) | Acetylene (C <sub>2</sub> H <sub>2</sub> ) | Ethylene (C <sub>2</sub> H <sub>4</sub> ) | Ethane (C <sub>2</sub> H <sub>6</sub> ) | Carbon Monoxide (CO) | Carbon Dioxide (CO <sub>2</sub> ) <sup>1</sup> | TDCG        |
|-------------|----------------------------|----------------------------|--------------------------------------------|-------------------------------------------|-----------------------------------------|----------------------|------------------------------------------------|-------------|
| Condition 1 | 100                        | 120                        | 35                                         | 50                                        | 65                                      | 350                  | 2,500                                          | 720         |
| Condition 2 | 101-700                    | 121-400                    | 36-50                                      | 51-100                                    | 66-100                                  | 351-570              | 2,500-4,000                                    | 721-1,920   |
| Condition 3 | 701-1,800                  | 401-1,000                  | 51-80                                      | 101-200                                   | 101-150                                 | 571-1,400            | 4,001-10,000                                   | 1,921-4,630 |
| Condition 4 | >1,800                     | >1,000                     | >80                                        | >200                                      | >150                                    | >1,400               | >10,000                                        | >4,630      |

IEEE C57.104-1991

**Table 1—Dissolved gas concentrations**

| Status      | Dissolved key gas concentration limits [ $\mu\text{L/L}$ (ppm) <sup>a</sup> ] |                            |                                            |                                           |                                         |                      |                                   |                   |
|-------------|-------------------------------------------------------------------------------|----------------------------|--------------------------------------------|-------------------------------------------|-----------------------------------------|----------------------|-----------------------------------|-------------------|
|             | Hydrogen (H <sub>2</sub> )                                                    | Methane (CH <sub>4</sub> ) | Acetylene (C <sub>2</sub> H <sub>2</sub> ) | Ethylene (C <sub>2</sub> H <sub>4</sub> ) | Ethane (C <sub>2</sub> H <sub>6</sub> ) | Carbon monoxide (CO) | Carbon dioxide (CO <sub>2</sub> ) | TDCG <sup>b</sup> |
| Condition 1 | 100                                                                           | 120                        | 1                                          | 50                                        | 65                                      | 350                  | 2 500                             | 720               |
| Condition 2 | 101–700                                                                       | 121–400                    | 2–9                                        | 51–100                                    | 66–100                                  | 351–570              | 2 500–4 000                       | 721–1920          |
| Condition 3 | 701–1800                                                                      | 401–1000                   | 10–35                                      | 101–200                                   | 101–150                                 | 571–1400             | 4 001–10 000                      | 1921–4630         |
| Condition 4 | >1800                                                                         | >1000                      | >35                                        | >200                                      | >150                                    | >1400                | >10 000                           | >4630             |

IEEE C57.104-2008

# Batasan Konsentrasi Gas IEEE C57.104-2019

- ▶ Table 1 dan Table 2 standard IEEE C57.104-2019 merupakan batasan untuk konsentrasi gas sebagai batasan konsentrasi gas pada table 1 standard IEEE C57.104-2008.
- ▶ Table 3 standard IEEE C57.104-2019 merupakan batasan dari Delta Gas, dimana Delta Gas sendiri merupakan kenaikan konsentrasi gas dari 2 pengujian terakhir.
- ▶ Table 4 standard IEEE C57.104-2019 merupakan batasan dari Rate Gas, dimana Rate Gas merupakan laju kenaikan dari gas dalam 1 tahun (ppm/tahun). Perhitungan dari Rate Gas memerlukan 3-6 data DGA dengan rentang waktu 4-24 bulan.

**Table 1—90<sup>th</sup> percentile gas concentrations as a function of O<sub>2</sub>/N<sub>2</sub> ratio and age in µL/L (ppm)**

|     |                                            | O <sub>2</sub> /N <sub>2</sub> Ratio ≤ 0.2 |       |         |     | O <sub>2</sub> /N <sub>2</sub> Ratio > 0.2 |       |         |     |
|-----|--------------------------------------------|--------------------------------------------|-------|---------|-----|--------------------------------------------|-------|---------|-----|
|     |                                            | Transformer Age in Years                   |       |         |     | Transformer Age in Years                   |       |         |     |
|     |                                            | Unknown                                    | 1 – 9 | 10 – 30 | >30 | Unknown                                    | 1 – 9 | 10 – 30 | >30 |
| Gas | Hydrogen (H <sub>2</sub> )                 | 80                                         | 75    |         | 100 | 40                                         | 40    |         |     |
|     | Methane (CH <sub>4</sub> )                 | 90                                         | 45    | 90      | 110 | 20                                         | 20    |         |     |
|     | Ethane (C <sub>2</sub> H <sub>6</sub> )    | 90                                         | 30    | 90      | 150 | 15                                         | 15    |         |     |
|     | Ethylene (C <sub>2</sub> H <sub>4</sub> )  | 50                                         | 20    | 50      | 90  | 50                                         | 25    | 60      |     |
|     | Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 1                                          | 1     |         |     | 2                                          | 2     |         |     |
|     | Carbon monoxide (CO)                       | 900                                        | 900   |         |     | 500                                        | 500   |         |     |
|     | Carbon dioxide (CO <sub>2</sub> )          | 9000                                       | 5000  | 10000   |     | 5000                                       | 3500  | 5500    |     |

NOTE—During the data analysis, it was determined that voltage class, MVA, and volume of mineral oil in the unit did not contribute in significant way to the determination of values provided in Table 1.

**Table 2—95<sup>th</sup> percentile gas concentrations as a function of O<sub>2</sub>/N<sub>2</sub> and age in µL/L (ppm)**

|     |                                            | O <sub>2</sub> /N <sub>2</sub> Ratio ≤ 0.2 |       |         |     | O <sub>2</sub> /N <sub>2</sub> Ratio > 0.2 |       |         |     |
|-----|--------------------------------------------|--------------------------------------------|-------|---------|-----|--------------------------------------------|-------|---------|-----|
|     |                                            | Transformer Age in Years                   |       |         |     | Transformer Age in Years                   |       |         |     |
|     |                                            | Unknown                                    | 1 – 9 | 10 – 30 | >30 | Unknown                                    | 1 – 9 | 10 – 30 | >30 |
| Gas | Hydrogen (H <sub>2</sub> )                 | 200                                        | 200   |         |     | 90                                         | 90    |         |     |
|     | Methane (CH <sub>4</sub> )                 | 150                                        | 100   | 150     | 200 | 50                                         | 60    |         | 30  |
|     | Ethane (C <sub>2</sub> H <sub>6</sub> )    | 175                                        | 70    | 175     | 250 | 40                                         | 30    | 40      |     |
|     | Ethylene (C <sub>2</sub> H <sub>4</sub> )  | 100                                        | 40    | 95      | 175 | 100                                        | 80    | 125     |     |
|     | Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 2                                          | 2     |         | 4   | 7                                          | 7     |         |     |
|     | Carbon monoxide (CO)                       | 1100                                       | 1100  |         |     | 600                                        | 600   |         |     |
|     | Carbon dioxide (CO <sub>2</sub> )          | 12500                                      | 7000  | 14000   |     | 7000                                       | 5000  | 8000    |     |

NOTE—During the data analysis, it was determined that voltage class, MVA, and volume of mineral oil in the unit did not contribute in significant way to the determination of values provided in Table 2

**Gassing Concentration <sup>1</sup>**  
**Table 1 & Table 2**

| <b>Low</b>       | <b>Medium</b>    | <b>High</b>      |
|------------------|------------------|------------------|
| <b>&lt; 40</b>   | <b>40-90</b>     | <b>&gt; 90</b>   |
| <b>&lt; 20</b>   | <b>20-60</b>     | <b>&gt; 60</b>   |
| <b>&lt; 15</b>   | <b>15-30</b>     | <b>&gt; 30</b>   |
| <b>&lt; 25</b>   | <b>25-80</b>     | <b>&gt; 80</b>   |
| <b>&lt; 2</b>    | <b>2-7</b>       | <b>&gt; 7</b>    |
| <b>&lt; 500</b>  | <b>500-600</b>   | <b>&gt; 600</b>  |
| <b>&lt; 3500</b> | <b>3500-5000</b> | <b>&gt; 5000</b> |

**Table 3— 95<sup>th</sup> percentile values for absolute level change between successive laboratory DGA samples in  $\mu\text{L/L}$  (ppm)**

|     |                                      | Maximum $\mu\text{L/L}$ (ppm) variation between consecutive laboratory DGA samples |                                       |
|-----|--------------------------------------|------------------------------------------------------------------------------------|---------------------------------------|
|     |                                      | $\text{O}_2/\text{N}_2$ Ratio $\leq 0.2$                                           | $\text{O}_2/\text{N}_2$ Ratio $> 0.2$ |
| Gas | Hydrogen ( $\text{H}_2$ )            | 40                                                                                 | 25                                    |
|     | Methane ( $\text{CH}_4$ )            | 30                                                                                 | 10                                    |
|     | Ethane ( $\text{C}_2\text{H}_6$ )    | 25                                                                                 | 7                                     |
|     | Ethylene ( $\text{C}_2\text{H}_4$ )  | 20                                                                                 |                                       |
|     | Acetylene ( $\text{C}_2\text{H}_2$ ) | Any Increase                                                                       |                                       |
|     | Carbon monoxide ( $\text{CO}$ )      | 250                                                                                | 175                                   |
|     | Carbon dioxide ( $\text{CO}_2$ )     | 2500                                                                               | 1750                                  |

NOTE—Contribution of voltage class, MVA, and volume of mineral oil in the unit was not studied for Table 3 as they have not been retained for Table 1 and Table 2. Data was insufficient to study age influence.

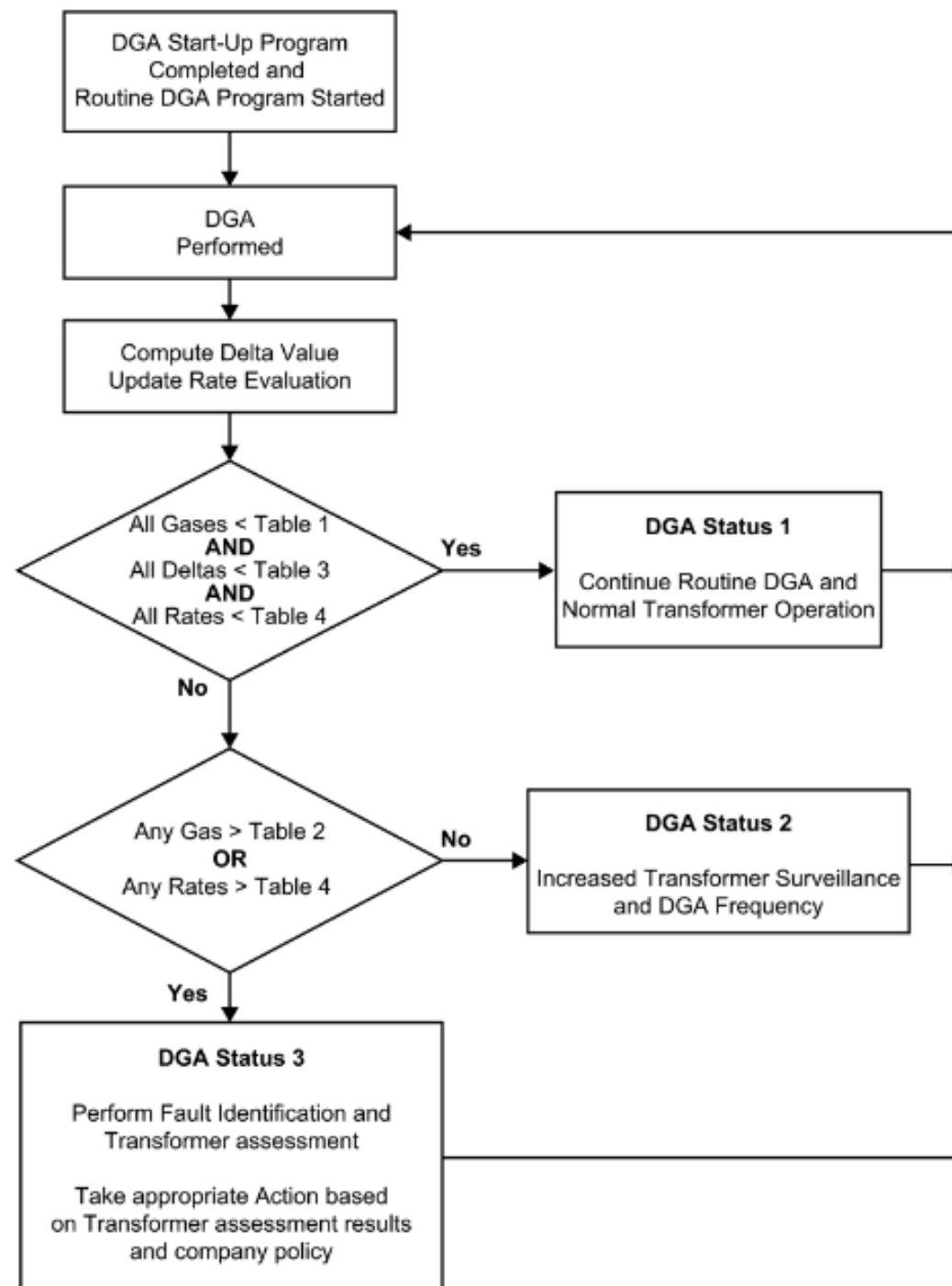
**Table 4—95<sup>th</sup> percentile values from multi-points (3-6 points) rate analysis of laboratory DGA samples with all gas levels below Table 1 values, in  $\mu\text{L/L/year}$  (ppm/year)**

|     |                                      | Maximum $\mu\text{L/L/year}$ (ppm/year) rate in function of the period between first and last point of the laboratory DGA series (3 to 6 samples) |              |                                       |              |
|-----|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------|--------------|
|     |                                      | $\text{O}_2/\text{N}_2$ Ratio $\leq 0.2$                                                                                                          |              | $\text{O}_2/\text{N}_2$ Ratio $> 0.2$ |              |
|     |                                      | Period between first and last point of the series                                                                                                 |              |                                       |              |
|     |                                      | 4-9 Months                                                                                                                                        | 10-24 Months | 4-9 Months                            | 10-24 Months |
| Gas | Hydrogen ( $\text{H}_2$ )            | 50                                                                                                                                                | 20           | 25                                    | 10           |
|     | Methane ( $\text{CH}_4$ )            | 15                                                                                                                                                | 10           | 4                                     | 3            |
|     | Ethane ( $\text{C}_2\text{H}_6$ )    | 15                                                                                                                                                | 9            | 3                                     | 2            |
|     | Ethylene ( $\text{C}_2\text{H}_4$ )  | 10                                                                                                                                                | 7            | 7                                     | 5            |
|     | Acetylene ( $\text{C}_2\text{H}_2$ ) | Any increasing rate                                                                                                                               |              | Any increasing rate                   |              |
|     | Carbon monoxide ( $\text{CO}$ )      | 200                                                                                                                                               | 100          | 100                                   | 80           |
|     | Carbon dioxide ( $\text{CO}_2$ )     | 1750                                                                                                                                              | 1000         | 1000                                  | 800          |

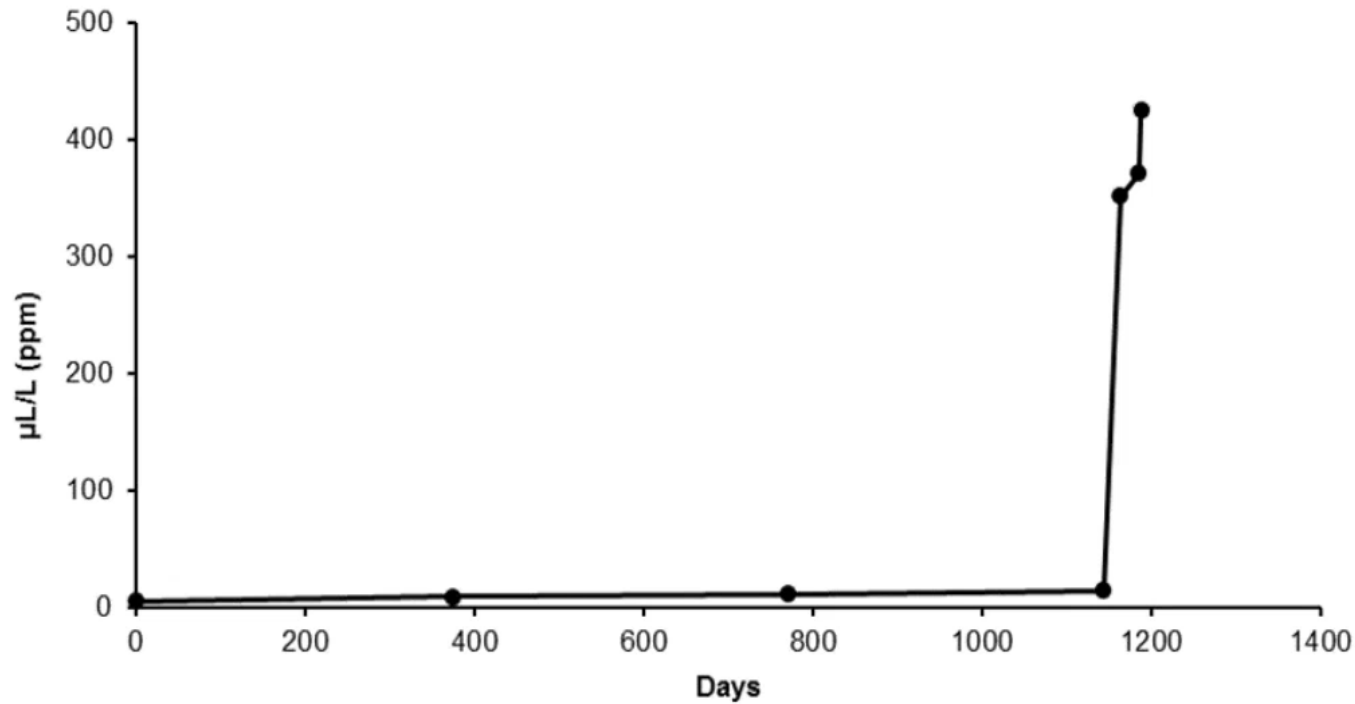
NOTE—Contribution of voltage class, MVA, and volume of mineral oil in the unit was not studied for Table 4 as they have not been retained for Table 1 and Table 2. Data was insufficient to study age influence.

### DELTA & RATE GAS CALCULATION

| Gas Parameter                             | Unit | Delta Gas <sup>2</sup> | Limit<br>Table 3 | Interval Pengujian:<br>(Bulan) 17 |                                 |                  |
|-------------------------------------------|------|------------------------|------------------|-----------------------------------|---------------------------------|------------------|
|                                           |      |                        |                  | Rate Gas<br>(Day)                 | Rate Gas <sup>3</sup><br>(Year) | Limit<br>Table 4 |
| Hydrogen (H <sub>2</sub> )                | ppm  | 5                      | 25               | 0.0104                            | 3.8                             | 10               |
| Methane (CH <sub>4</sub> )                | ppm  | 1                      | 10               | -0.0015                           | -0.5                            | 3                |
| Ethane (C <sub>2</sub> H <sub>6</sub> )   | ppm  | 0                      | 7                | 0                                 | 0.0                             | 2                |
| Ethylene (C <sub>2</sub> H <sub>4</sub> ) | ppm  | 0                      | 20               | 0                                 | 0.0                             | 5                |
| Acetylene(C <sub>2</sub> H <sub>2</sub> ) | ppm  | 0                      | Any              | 0                                 | 0.0                             | Any              |
| Carbon Monoxide (CO)                      | ppm  | 16                     | 175              | 0.0724                            | 26.4                            | 80               |
| Carbon Dioxide (CO <sub>2</sub> )         | ppm  | 108                    | 1750             | 0.484                             | 176.7                           | 800              |



# Contoh 1: Hasil tanpa perlu tabel



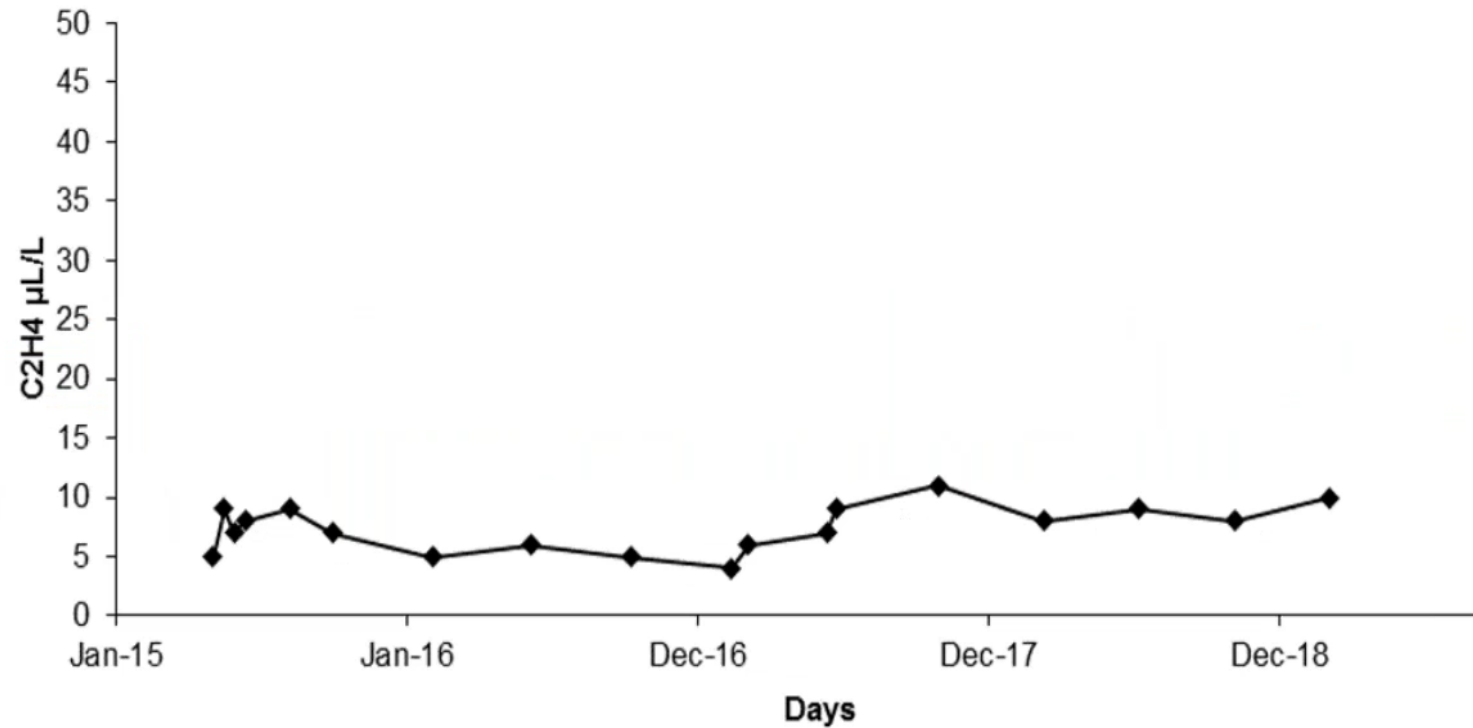
Action: Konsultasi pada expert DGA dan investigasi

## Contoh 2

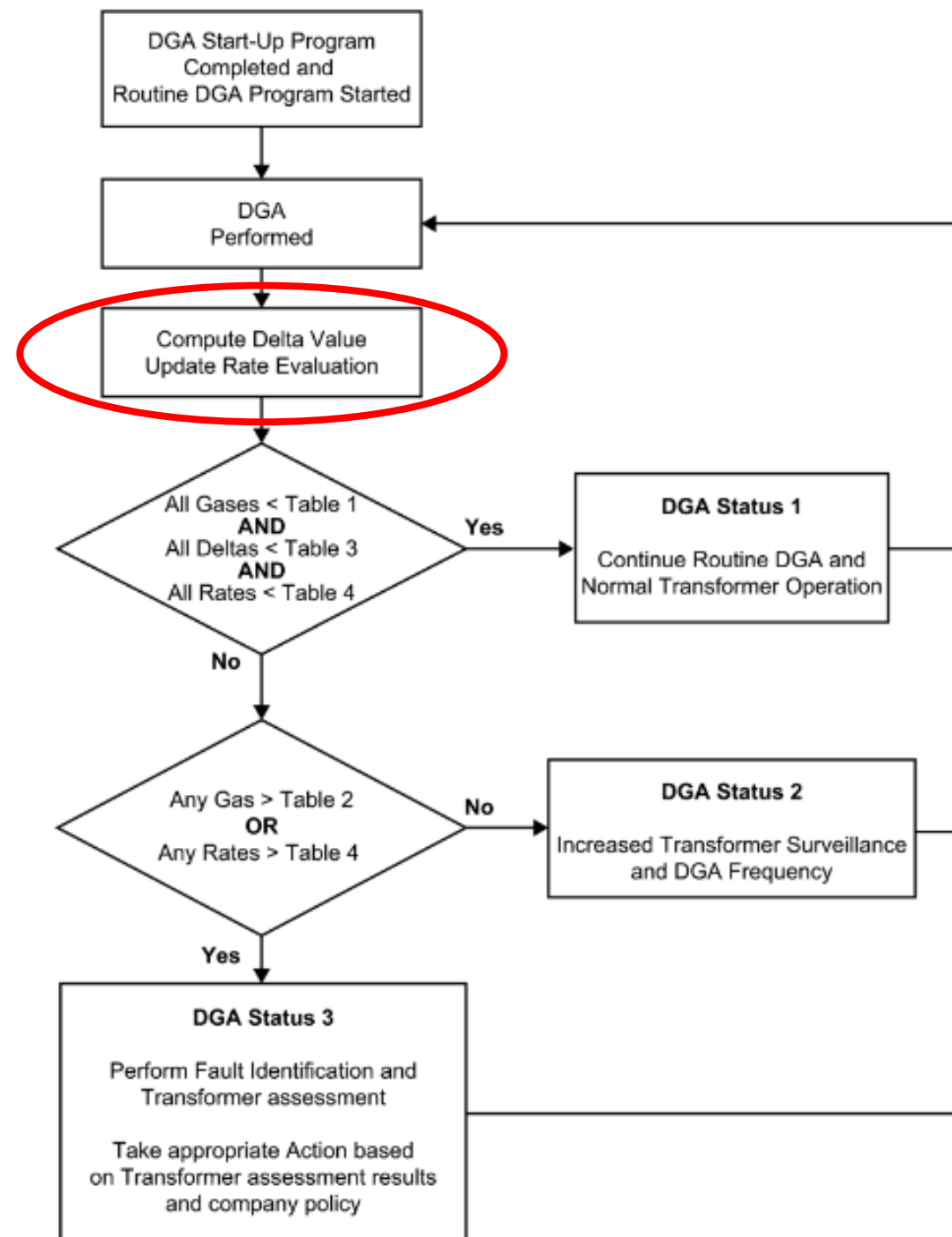
|                                             | 3/4/19 | 11/4/18 | 7/7/18 | 3/9/18 | 10/28/17 | 6/23/17 |
|---------------------------------------------|--------|---------|--------|--------|----------|---------|
| Hydrogen (H <sub>2</sub> )                  | 35     | 28      | 31     | 34     | 29       | 32      |
| Methane (CH <sub>4</sub> )                  | 3      | 5       | 2      | 4      | 5        | 4       |
| Ethane (C <sub>2</sub> H <sub>6</sub> )     | 4      | 5       | 4      | 6      | 5        | 5       |
| Etylene (C <sub>2</sub> H <sub>4</sub> )    | 10     | 8       | 9      | 8      | 11       | 9       |
| Acethylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 0       | 0      | 0      | 0        | 0       |
| Carbon Monoxyde (CO)                        | 351    | 324     | 364    | 332    | 341      | 349     |
| Carbon Dioxyde (CO <sub>2</sub> )           | 3452   | 3567    | 3487   | 3501   | 3525     | 3475    |
| Oxygen (O <sub>2</sub> )                    | 4002   | 3890    | 4200   | 4289   | 4421     | 4054    |
| Nitrogen (N <sub>2</sub> )                  | 81952  | 85341   | 84773  | 83990  | 85143    | 82154   |

Transformator dengan usia 12 tahun

# Grafik Contoh Data



Dari grafik tidak ada indikasi masalah, selanjutnya cek dengan table



## Contoh 2: Menghitung Delta Gas

|                                            | 3/4/19 | 11/4/18 | 7/7/18 | 3/9/18 | 10/28/17 | 6/23/17 |
|--------------------------------------------|--------|---------|--------|--------|----------|---------|
| Hydrogen (H <sub>2</sub> )                 | 35     | 28      | 31     | 34     | 29       | 32      |
| Methane (CH <sub>4</sub> )                 | 3      | 5       | 2      | 4      | 5        | 4       |
| Ethane (C <sub>2</sub> H <sub>6</sub> )    | 4      | 5       | 4      | 6      | 5        | 5       |
| Etylene (C <sub>2</sub> H <sub>4</sub> )   | 10     | 8       | 9      | 11     | 8        | 9       |
| Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 0       | 0      | 0      | 0        | 0       |
| Carbon Monoxyde (CO)                       | 351    | 324     | 364    | 332    | 341      | 349     |
| Carbon Dioxide (CO <sub>2</sub> )          | 3452   | 3567    | 3487   | 3501   | 3525     | 3475    |
| Oxygen (O <sub>2</sub> )                   | 4002   | 3890    | 4200   | 4289   | 4421     | 4054    |
| Nitrogen (N <sub>2</sub> )                 | 81952  | 85341   | 84773  | 83990  | 85143    | 82154   |

$$\text{Delta} = 3/4/19 - 11/4/18$$

## Contoh 2: Menghitung Delta Gas

|                                            | 3/4/19 | 11/4/18 | Delta |
|--------------------------------------------|--------|---------|-------|
| Hydrogen (H <sub>2</sub> )                 | 35     | 28      | 7     |
| Methane (CH <sub>4</sub> )                 | 3      | 5       | -2    |
| Ethane (C <sub>2</sub> H <sub>6</sub> )    | 4      | 5       | -1    |
| Etylene (C <sub>2</sub> H <sub>4</sub> )   | 10     | 8       | 2     |
| Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 0       | 0     |
| Carbon Monoxyde (CO)                       | 351    | 324     | 27    |
| Carbon Dioxyme (CO <sub>2</sub> )          | 3452   | 3567    | -115  |
| Oxygen (O <sub>2</sub> )                   | 4002   | 3890    |       |
| Nitrogen (N <sub>2</sub> )                 | 81952  | 85341   |       |

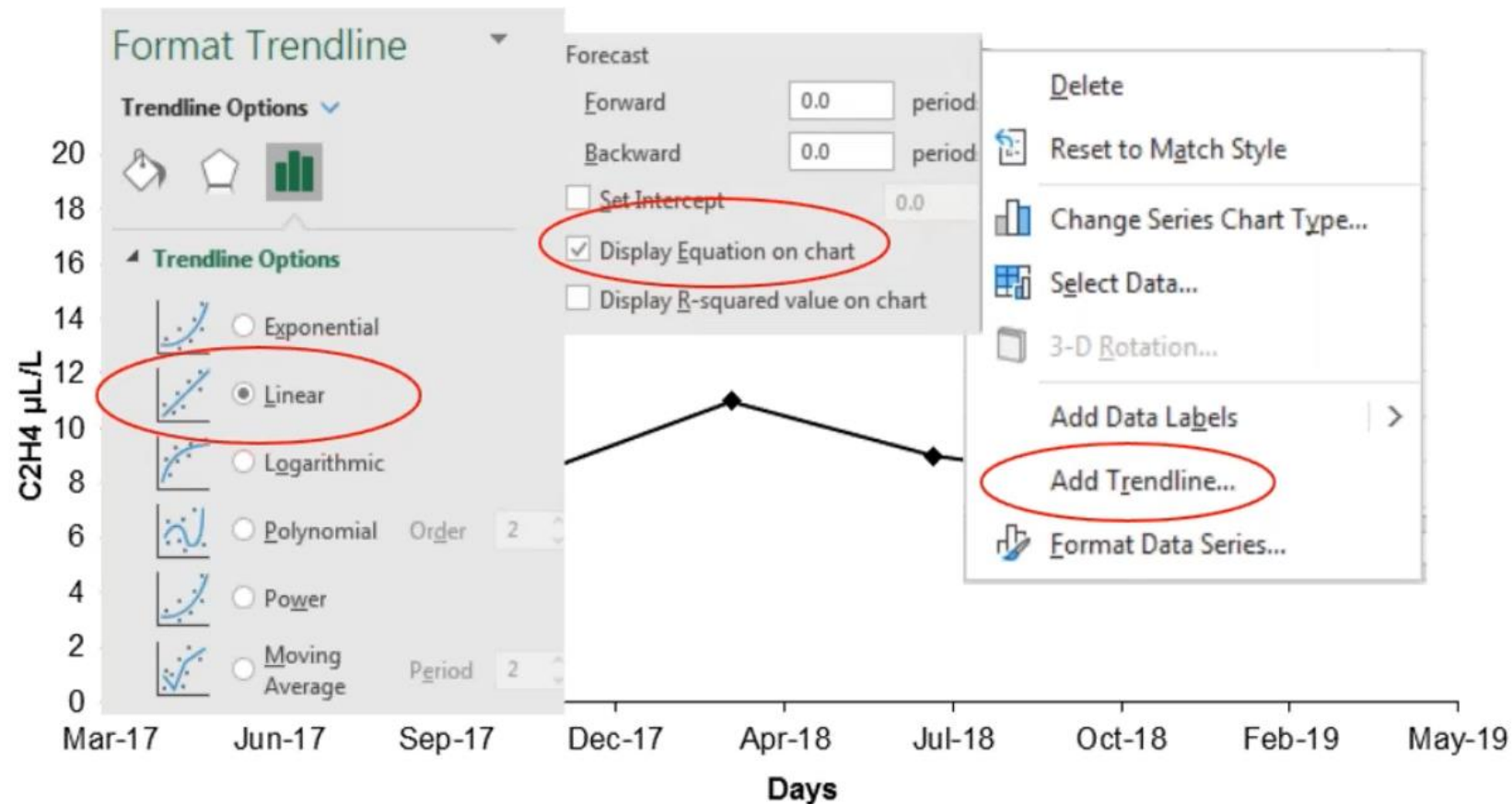
Delta in ul/l = 3/4/19 – 11/4/18

## Contoh 2: Menghitung rate ppm/tahun

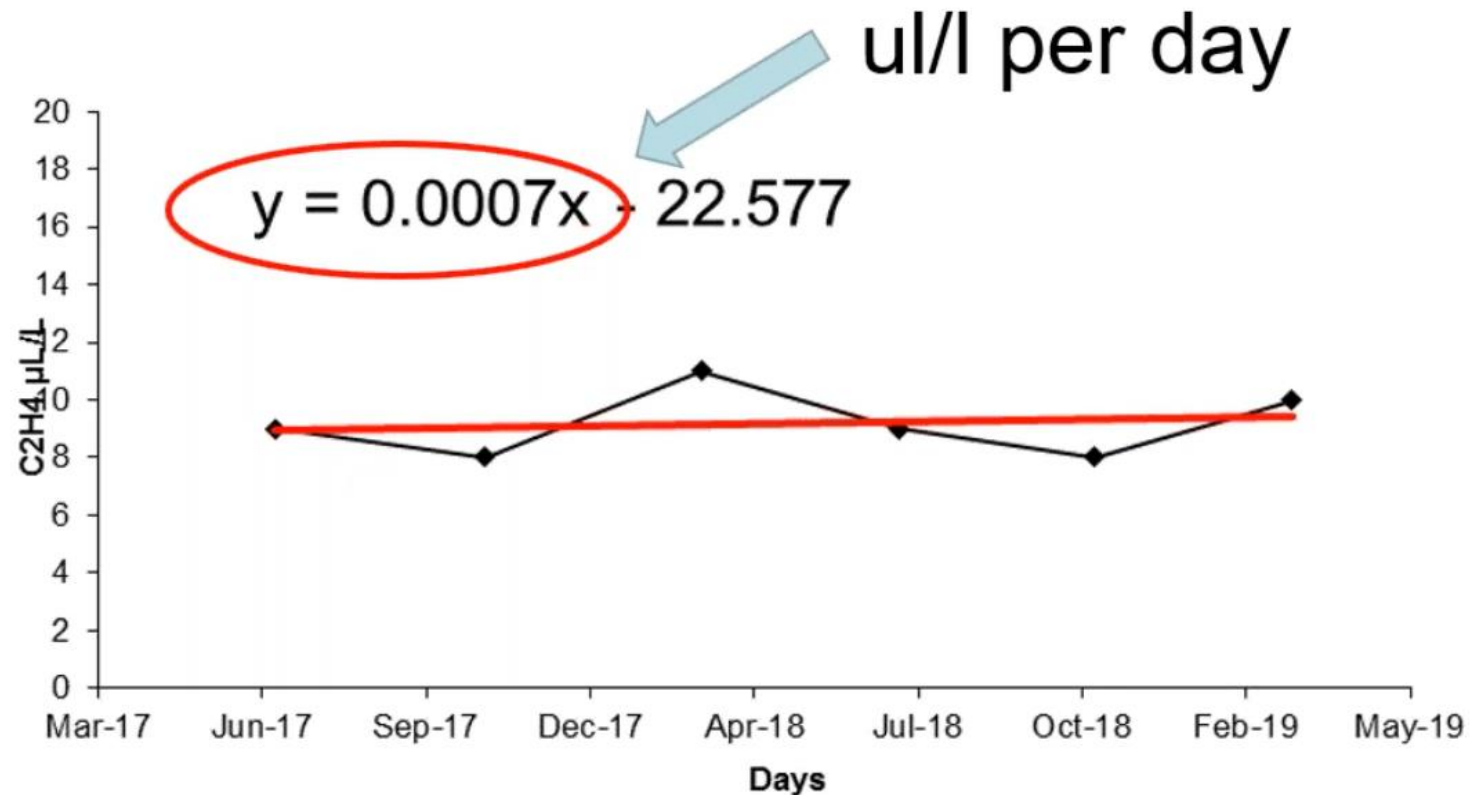
|                                            | 3/4/19 | 11/4/18 | 7/7/18 | 3/9/18 | 10/28/17 | 6/23/17 |
|--------------------------------------------|--------|---------|--------|--------|----------|---------|
| Hydrogen (H <sub>2</sub> )                 | 35     | 28      | 31     | 34     | 29       | 32      |
| Methane (CH <sub>4</sub> )                 | 3      | 5       | 2      | 4      | 5        | 4       |
| Ethane (C <sub>2</sub> H <sub>6</sub> )    | 4      | 5       | 4      | 6      | 5        | 5       |
| Etylene (C <sub>2</sub> H <sub>4</sub> )   | 10     | 8       | 9      | 11     | 8        | 9       |
| Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 0       | 0      | 0      | 0        | 0       |
| Carbon Monoxyde (CO)                       | 351    | 324     | 364    | 332    | 341      | 349     |
| Carbon Dioxyde (CO <sub>2</sub> )          | 3452   | 3567    | 3487   | 3501   | 3525     | 3475    |
| Oxygen (O <sub>2</sub> )                   | 4002   | 3890    | 4200   | 4289   | 4421     | 4054    |
| Nitrogen (N <sub>2</sub> )                 | 81952  | 85341   | 84773  | 83990  | 85143    | 82154   |

Rate diambil dari 3-6 data dengan batas waktu 4-24 Bulan

## Contoh 2: Rate bisa dihitung dengan grafik excel (mX+b)



## Contoh 2

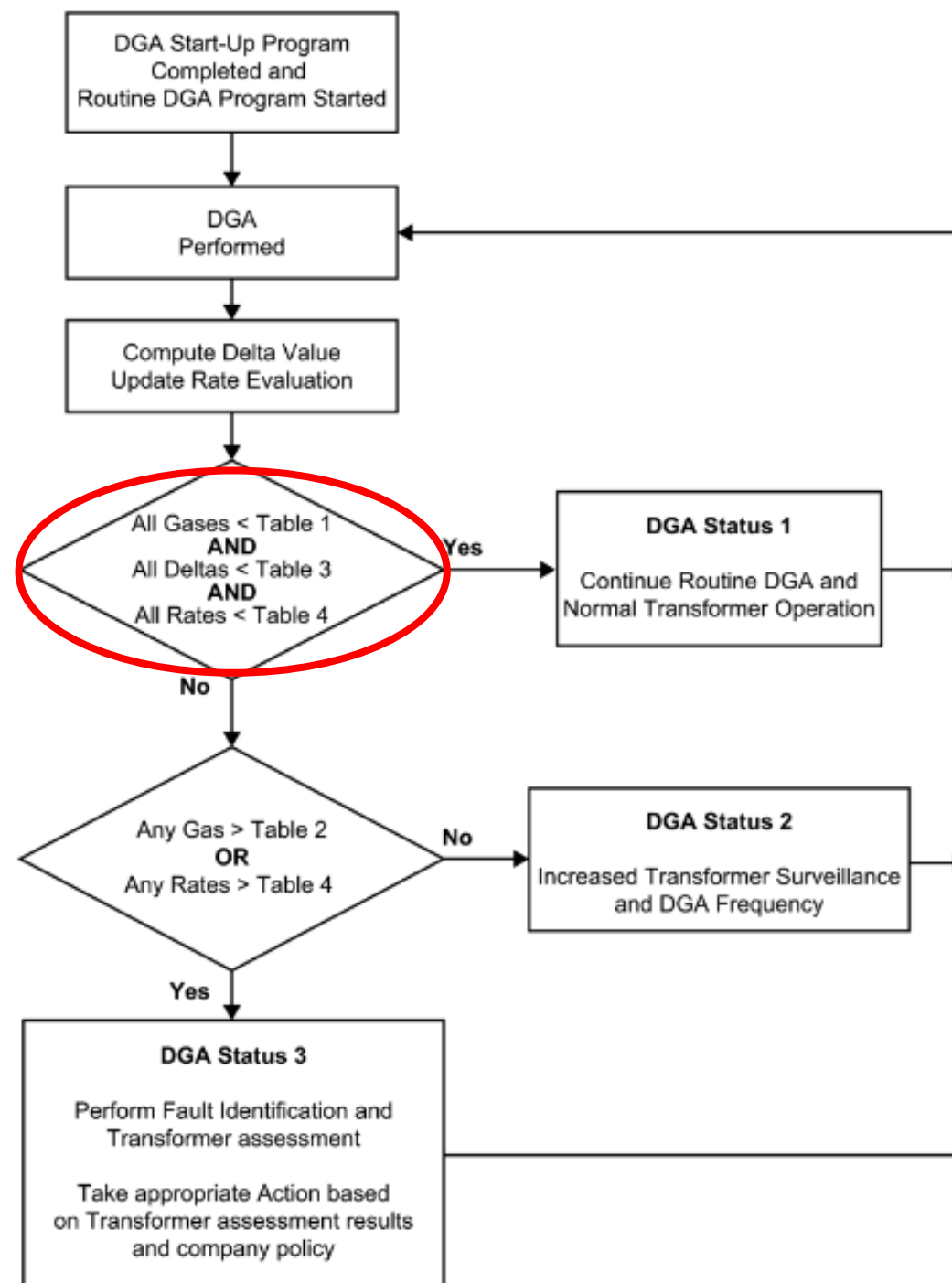


$$\text{Rate} = 0.0007 * 365 = 0.3 \text{ ul/l/year}$$

## Contoh 2: Menghitung Rasio O<sub>2</sub>/N<sub>2</sub>

|                                            | 3/4/19 | 11/4/18 | 7/7/18 | 3/9/18 | 10/28/17 | 6/23/17 |
|--------------------------------------------|--------|---------|--------|--------|----------|---------|
| Hydrogen (H <sub>2</sub> )                 | 35     | 28      | 31     | 34     | 29       | 32      |
| Methane (CH <sub>4</sub> )                 | 3      | 5       | 2      | 4      | 5        | 4       |
| Ethane (C <sub>2</sub> H <sub>6</sub> )    | 4      | 5       | 4      | 6      | 5        | 5       |
| Etylene (C <sub>2</sub> H <sub>4</sub> )   | 10     | 8       | 9      | 11     | 8        | 9       |
| Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 0       | 0      | 0      | 0        | 0       |
| Carbon Monoxide (CO)                       | 351    | 324     | 364    | 332    | 341      | 349     |
| Carbon Dioxide (CO <sub>2</sub> )          | 3452   | 3567    | 3487   | 3501   | 3525     | 3475    |
| Oxygen (O <sub>2</sub> )                   | 4002   | 3890    | 4200   | 4289   | 4421     | 4054    |
| Nitrogen (N <sub>2</sub> )                 | 81952  | 85341   | 84773  | 83990  | 85143    | 82154   |

$$\text{O}_2/\text{N}_2 \text{ Ratio} = 4002 / 81952 = 0.048$$



## Contoh 2: Perbandingan Table 1

Table 1—90<sup>th</sup> percentile gas concentrations as a function of O<sub>2</sub>/N<sub>2</sub> ratio and age in  $\mu\text{L/L}$  (ppm)

|                                                                                                                                                                                                           |                                            | O <sub>2</sub> /N <sub>2</sub> Ratio $\leq 0.2$ |       |         |     | O <sub>2</sub> /N <sub>2</sub> Ratio $> 0.2$ |       |         |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------|-------|---------|-----|----------------------------------------------|-------|---------|------|
|                                                                                                                                                                                                           |                                            | Transformer Age in Years                        |       |         |     | Transformer Age in Years                     |       |         |      |
|                                                                                                                                                                                                           |                                            | Unknown                                         | 1 – 9 | 10 – 30 | >30 | Unknown                                      | 1 – 9 | 10 – 30 | >30  |
| Gas                                                                                                                                                                                                       | Hydrogen (H <sub>2</sub> )                 | 80                                              |       | 75      | 100 | 40                                           |       | 40      |      |
|                                                                                                                                                                                                           | Methane (CH <sub>4</sub> )                 | 90                                              | 45    | 90      | 110 | 20                                           |       | 20      |      |
|                                                                                                                                                                                                           | Ethane (C <sub>2</sub> H <sub>6</sub> )    | 90                                              | 30    | 90      | 150 | 15                                           |       | 15      |      |
|                                                                                                                                                                                                           | Ethylene (C <sub>2</sub> H <sub>4</sub> )  | 50                                              | 20    | 50      | 90  | 50                                           | 25    |         | 60   |
|                                                                                                                                                                                                           | Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 1                                               |       | 1       |     | 2                                            |       | 2       |      |
|                                                                                                                                                                                                           | Carbon monoxide (CO)                       | 900                                             |       | 900     |     | 500                                          |       | 500     |      |
|                                                                                                                                                                                                           | Carbon dioxide (CO <sub>2</sub> )          | 9000                                            | 5000  | 10000   |     | 5000                                         | 3500  |         | 5500 |
| NOTE—During the data analysis, it was determined that voltage class, MVA, and volume of mineral oil in the unit did not contribute in significant way to the determination of values provided in Table 1. |                                            |                                                 |       |         |     |                                              |       |         |      |

O<sub>2</sub>/N<sub>2</sub> Ratio = 0.048 & Age = 12 years

## Contoh 2: Perbandingan Table 3

Table 3— 95<sup>th</sup> percentile values for absolute level change between successive laboratory DGA samples in  $\mu\text{L/L}$  (ppm)

|                                                                                                                                                                                                                  |                                      | Maximum $\mu\text{L/L}$ (ppm) variation between consecutive laboratory DGA samples |                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------|---------------------------------------|
|                                                                                                                                                                                                                  |                                      | $\text{O}_2/\text{N}_2$ Ratio $\leq 0.2$                                           | $\text{O}_2/\text{N}_2$ Ratio $> 0.2$ |
| Gas                                                                                                                                                                                                              | Hydrogen ( $\text{H}_2$ )            | 40                                                                                 | 25                                    |
|                                                                                                                                                                                                                  | Methane ( $\text{CH}_4$ )            | 30                                                                                 | 10                                    |
|                                                                                                                                                                                                                  | Ethane ( $\text{C}_2\text{H}_6$ )    | 25                                                                                 | 7                                     |
|                                                                                                                                                                                                                  | Ethylene ( $\text{C}_2\text{H}_4$ )  | 20                                                                                 | Any Increase                          |
|                                                                                                                                                                                                                  | Acetylene ( $\text{C}_2\text{H}_2$ ) | 20                                                                                 |                                       |
|                                                                                                                                                                                                                  | Carbon monoxide ( $\text{CO}$ )      | 250                                                                                | 175                                   |
|                                                                                                                                                                                                                  | Carbon dioxide ( $\text{CO}_2$ )     | 2500                                                                               | 1750                                  |
| NOTE—Contribution of voltage class, MVA, and volume of mineral oil in the unit was not studied for Table 3 as they have not been retained for Table 1 and Table 2. Data was insufficient to study age influence. |                                      |                                                                                    |                                       |

$$\text{O}_2/\text{N}_2 \text{ Ratio} = 0.048$$

# Contoh 2: Perbandingan Table 4

**Table 4—95<sup>th</sup> percentile values from multi-points (3-6 points) rate analysis of laboratory DGA samples with all gas levels below Table 1 values, in  $\mu\text{L/L/year}$  (ppm/year)**

|     |                                      | Maximum $\mu\text{L/L/year}$ (ppm/year) rate in function of the period between first and last point of the laboratory DGA series (3 to 6 samples) |              |                                       |              |
|-----|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------|--------------|
|     |                                      | $\text{O}_2/\text{N}_2$ Ratio $\leq 0.2$                                                                                                          |              | $\text{O}_2/\text{N}_2$ Ratio $> 0.2$ |              |
|     |                                      | Period between first and last point of the series                                                                                                 |              |                                       |              |
|     |                                      | 4-9 Months                                                                                                                                        | 10-24 Months | 4-9 Months                            | 10-24 Months |
| Gas | Hydrogen ( $\text{H}_2$ )            | 50                                                                                                                                                | 20           | 25                                    | 10           |
|     | Methane ( $\text{CH}_4$ )            | 15                                                                                                                                                | 10           | 4                                     | 3            |
|     | Ethane ( $\text{C}_2\text{H}_6$ )    | 15                                                                                                                                                | 9            | 3                                     | 2            |
|     | Ethylene ( $\text{C}_2\text{H}_4$ )  | 10                                                                                                                                                | 7            | 7                                     | 5            |
|     | Acetylene ( $\text{C}_2\text{H}_2$ ) | Any increasing rate                                                                                                                               |              | Any increasing rate                   |              |
|     | Carbon monoxide ( $\text{CO}$ )      | 200                                                                                                                                               | 100          | 100                                   | 80           |
|     | Carbon dioxide ( $\text{CO}_2$ )     | 1750                                                                                                                                              | 1000         | 1000                                  | 800          |

NOTE—Contribution of voltage class, MVA, and volume of mineral oil in the unit was not studied for Table 4 as they have not been retained for Table 1 and Table 2. Data was insufficient to study age influence.

## Contoh 2: Hasil Perbandingan

<=?

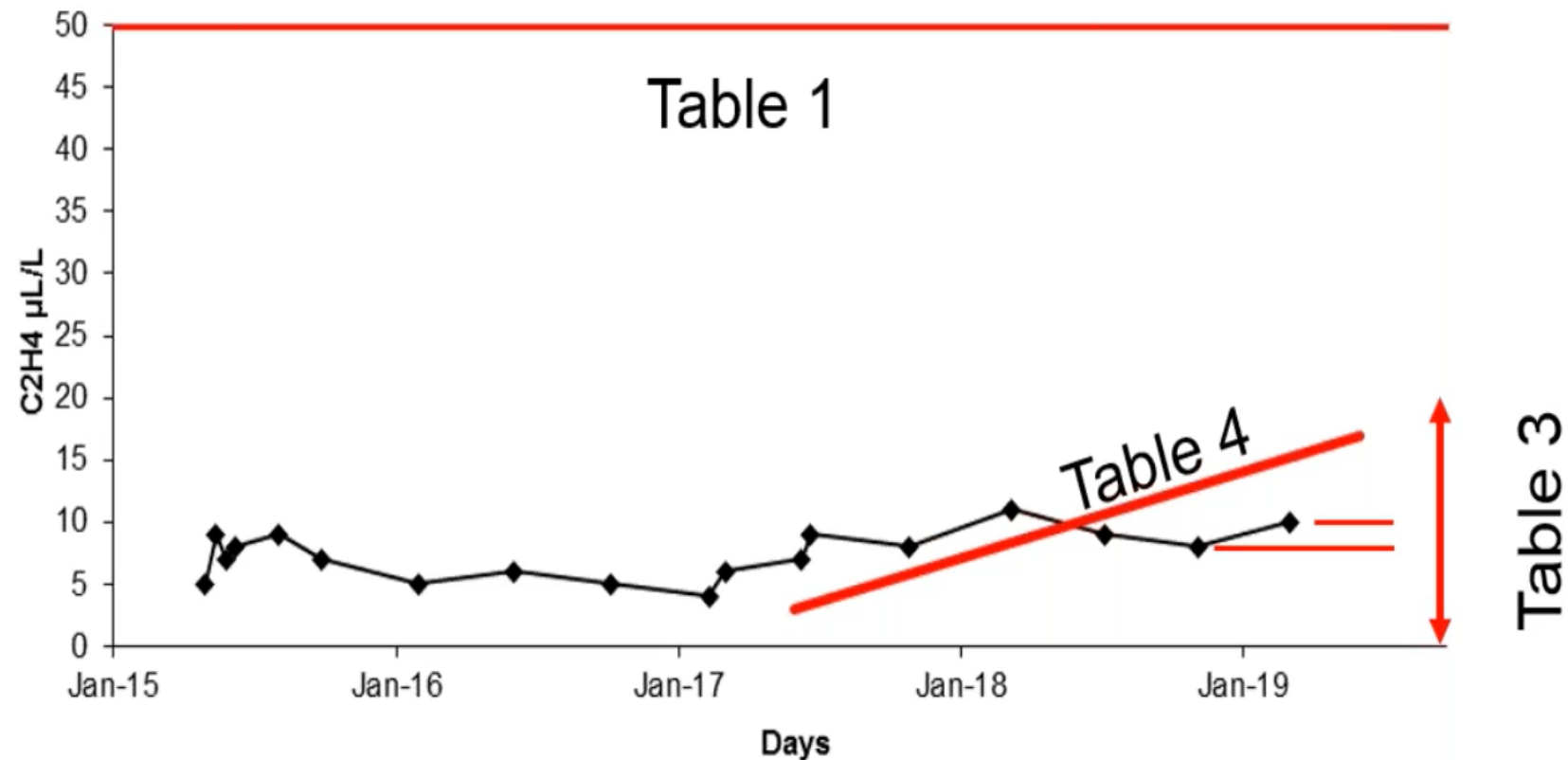
<=?

<=?

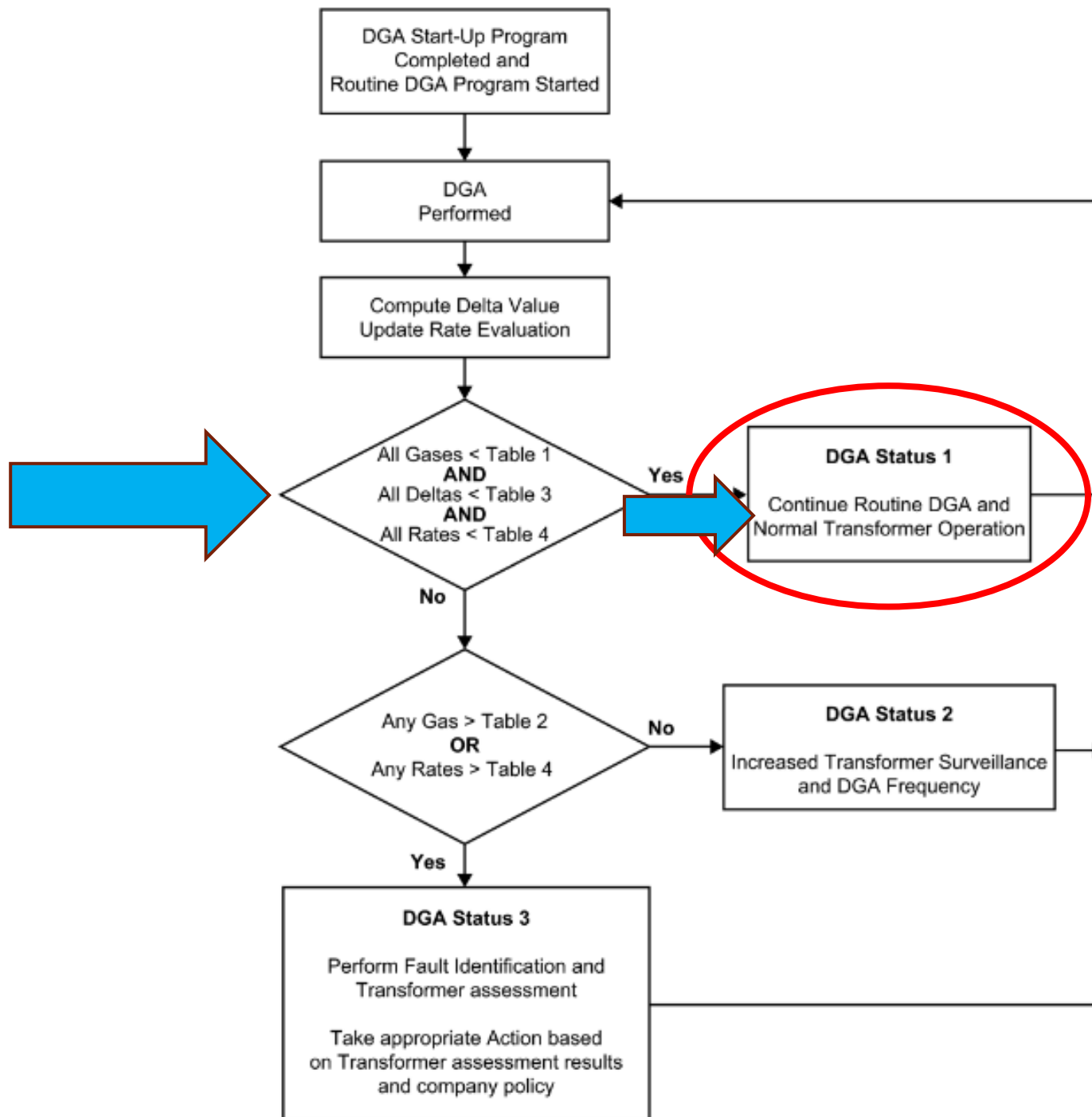
|                                             | 3/4/19 | Table 1 | Delta | Table 3 | Rate | Table 4 |
|---------------------------------------------|--------|---------|-------|---------|------|---------|
| Hydrogen (H <sub>2</sub> )                  | 35     | 75      | 7     | 40      | 0.8  | 20      |
| Methane (CH <sub>4</sub> )                  | 3      | 90      | -2    | 30      | -0.6 | 10      |
| Ethane (C <sub>2</sub> H <sub>6</sub> )     | 4      | 90      | -1    | 25      | -0.6 | 9       |
| Etylene (C <sub>2</sub> H <sub>4</sub> )    | 10     | 50      | 2     | 20      | 0.3  | 7       |
| Acethylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 1       | 0     | >0      | 0.0  | >0      |
| Carbon Monoxyde (CO)                        | 351    | 900     | 27    | 250     | -0.8 | 100     |
| Carbon Dioxyde (CO <sub>2</sub> )           | 3452   | 10000   | -115  | 2500    | 0.0  | 1000    |
| Oxygen (O <sub>2</sub> )                    | 4002   |         |       |         |      |         |
| Nitrogen (N <sub>2</sub> )                  | 81952  |         |       |         |      |         |

DGA Status = Status 1

## Contoh 2: Continue Pengujian Rutin



Action: Continue routine testing



## Contoh 3: Ada perubahan signifikan

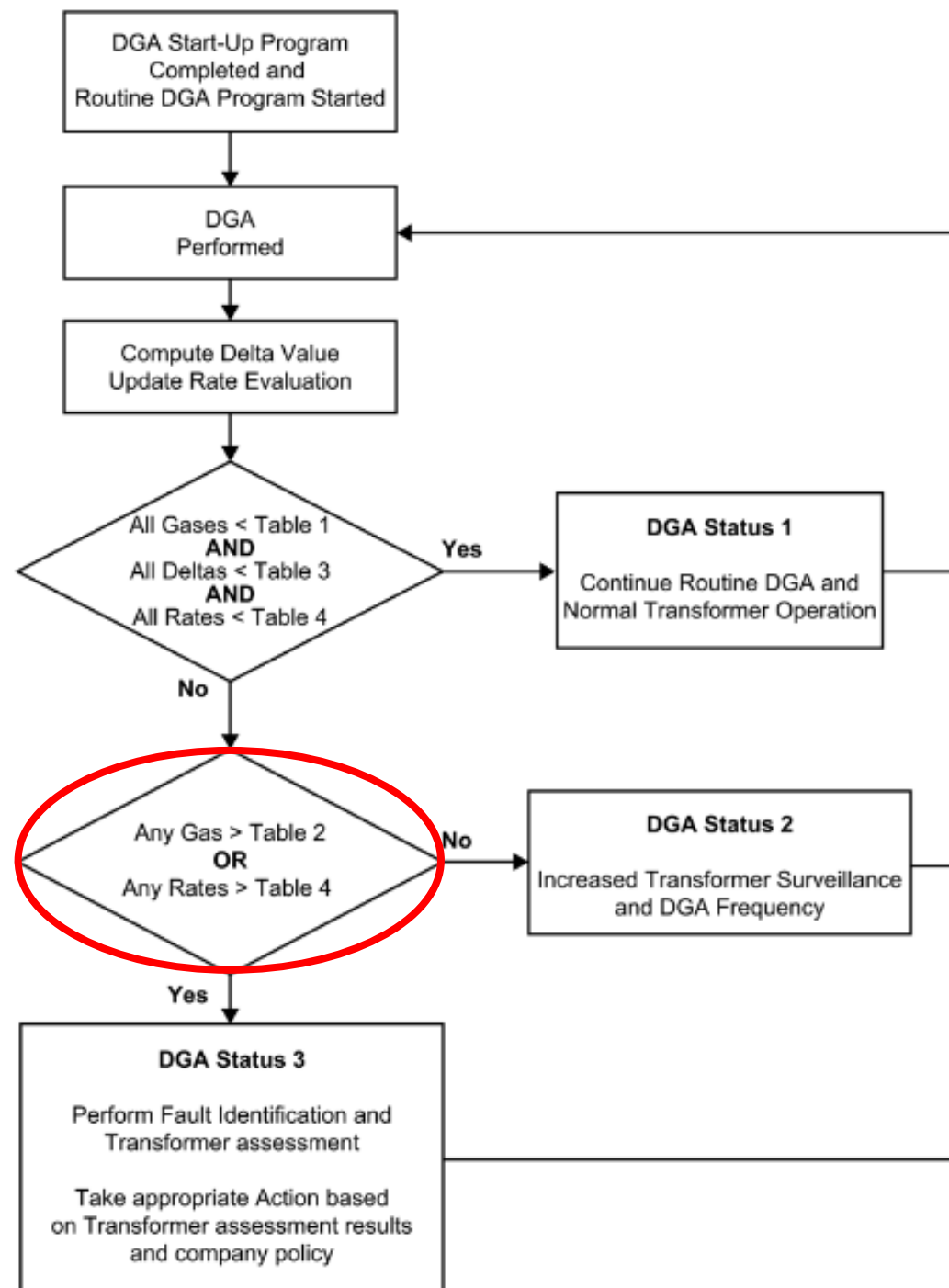
|                                             | 3/4/19 | 11/4/18 | Delta |
|---------------------------------------------|--------|---------|-------|
| Hydrogen (H <sub>2</sub> )                  | 35     | 28      | 7     |
| Methane (CH <sub>4</sub> )                  | 3      | 5       | -2    |
| Ethane (C <sub>2</sub> H <sub>6</sub> )     | 4      | 5       | -1    |
| Etylene (C <sub>2</sub> H <sub>4</sub> )    | 29     | 8       | 21    |
| Acethylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 0       | 0     |
| Carbon Monoxyde (CO)                        | 351    | 324     | 27    |
| Carbon Dioxyme (CO <sub>2</sub> )           | 3452   | 3567    | -115  |
| Oxygen (O <sub>2</sub> )                    | 4002   | 3890    |       |
| Nitrogen (N <sub>2</sub> )                  | 81952  | 85341   |       |

Delta in ul/l = 3/4/19 – 11/4/18

## Contoh 3: Perbandingan tabel

|                                            | 3/4/19 | Table 1 | Delta | Table 3 | Rate | Table 4 |
|--------------------------------------------|--------|---------|-------|---------|------|---------|
| Hydrogen (H <sub>2</sub> )                 | 35     | 75      | 7     | 40      | 0.8  | 20      |
| Methane (CH <sub>4</sub> )                 | 3      | 90      | -2    | 30      | -0.6 | 10      |
| Ethane (C <sub>2</sub> H <sub>6</sub> )    | 4      | 90      | -1    | 25      | -0.6 | 9       |
| Etylene (C <sub>2</sub> H <sub>4</sub> )   | 29     | 50      | 21    | 20      | 8.1  | 7       |
| Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 1       | 0     | >0      | 0.0  | >0      |
| Carbon Monoxyde (CO)                       | 351    | 900     | 27    | 250     | -0.8 | 100     |
| Carbon Dioxyde (CO <sub>2</sub> )          | 3452   | 10000   | -115  | 2500    | 0.0  | 1000    |
| Oxygen (O <sub>2</sub> )                   | 4002   |         |       |         |      |         |
| Nitrogen (N <sub>2</sub> )                 | 81952  |         |       |         |      |         |

DGA Status = Tidak pada Status 1



## Contoh 3: Perbandingan Table 2

## Table 2

**Table 2—95<sup>th</sup> percentile gas concentrations as a function of O<sub>2</sub>/N<sub>2</sub> and age in  $\mu\text{L/L}$  (ppm)**

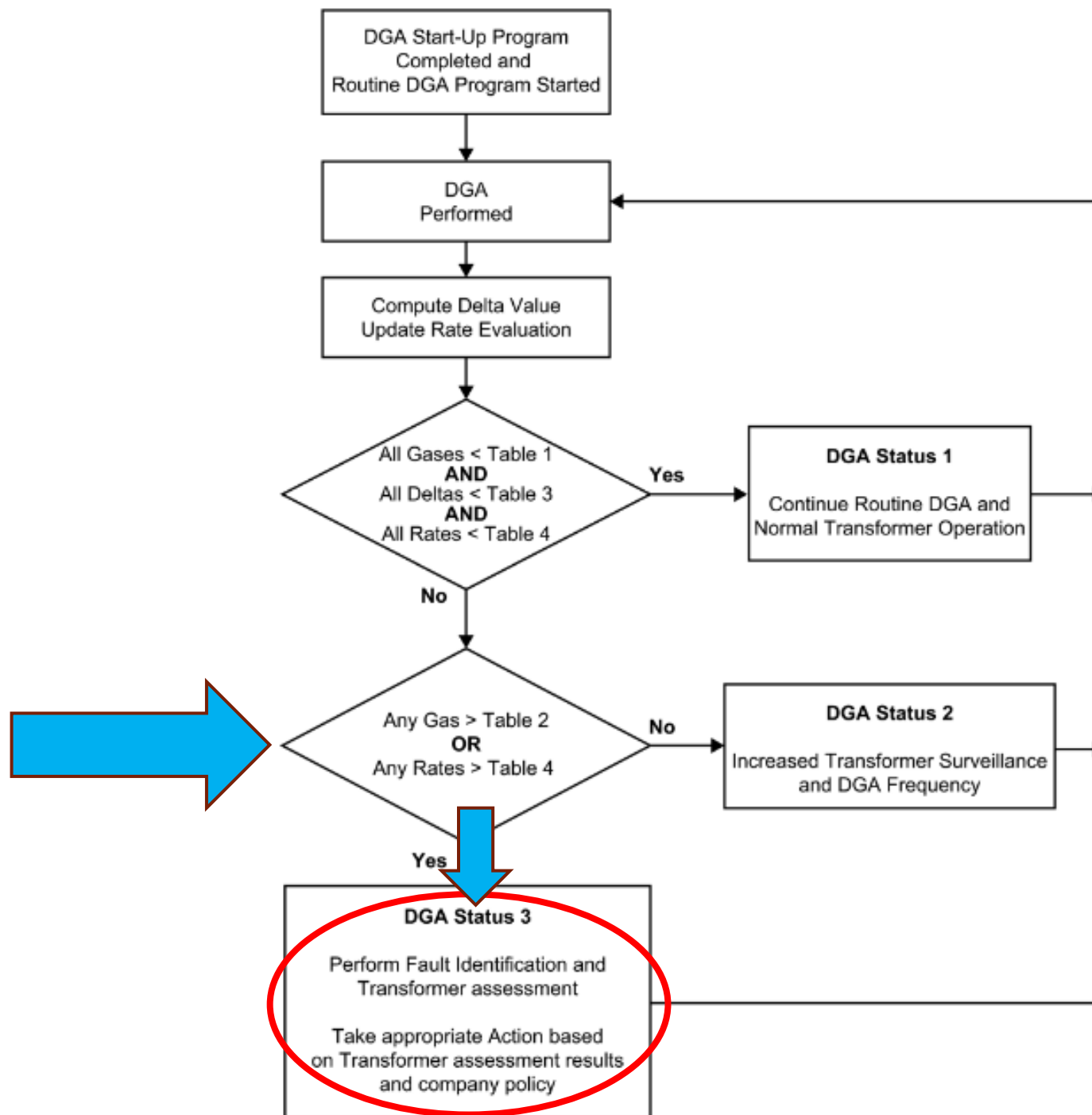
|            |                                               | <b>O<sub>2</sub>/N<sub>2</sub> Ratio ≤ 0.2</b> |              |                |               | <b>O<sub>2</sub>/N<sub>2</sub> Ratio &gt; 0.2</b> |              |                |               |
|------------|-----------------------------------------------|------------------------------------------------|--------------|----------------|---------------|---------------------------------------------------|--------------|----------------|---------------|
|            |                                               | <b>Transformer Age in Years</b>                |              |                |               | <b>Transformer Age in Years</b>                   |              |                |               |
|            |                                               | <b>Unknown</b>                                 | <b>1 – 9</b> | <b>10 – 30</b> | <b>&gt;30</b> | <b>Unknown</b>                                    | <b>1 – 9</b> | <b>10 – 30</b> | <b>&gt;30</b> |
| <b>Gas</b> | <b>Hydrogen (H<sub>2</sub>)</b>               | 200                                            |              | 200            |               | 90                                                | 90           |                |               |
|            | <b>Methane (CH<sub>4</sub>)</b>               | 150                                            | 100          | 150            | 200           | 50                                                | 60           |                | 30            |
|            | <b>Ethane (C<sub>2</sub>H<sub>6</sub>)</b>    | 175                                            | 70           | 175            | 250           | 40                                                | 30           | 40             |               |
|            | <b>Ethylene (C<sub>2</sub>H<sub>4</sub>)</b>  | 100                                            | 40           | 95             | 175           | 100                                               | 80           | 125            |               |
|            | <b>Acetylene (C<sub>2</sub>H<sub>2</sub>)</b> | 2                                              |              | 2              | 4             | 7                                                 | 7            |                |               |
|            | <b>Carbon monoxide (CO)</b>                   | 1100                                           |              | 1100           |               | 600                                               | 600          |                |               |
|            | <b>Carbon dioxide (CO<sub>2</sub>)</b>        | 12500                                          | 7000         | 14000          |               | 7000                                              | 5000         | 8000           |               |

NOTE—During the data analysis, it was determined that voltage class, MVA, and volume of mineral oil in the unit did not contribute in significant way to the determination of values provided in Table 2

## Contoh 3: Perbandingan Tabel

|                                            |        |         |  |  |      |         |
|--------------------------------------------|--------|---------|--|--|------|---------|
|                                            | >?     |         |  |  | >?   |         |
|                                            | 3/4/19 | Table 2 |  |  | Rate | Table 4 |
| Hydrogen (H <sub>2</sub> )                 | 35     | 200     |  |  | 0.8  | 20      |
| Methane (CH <sub>4</sub> )                 | 3      | 150     |  |  | -0.6 | 10      |
| Ethane (C <sub>2</sub> H <sub>6</sub> )    | 4      | 175     |  |  | -0.6 | 9       |
| Etylene (C <sub>2</sub> H <sub>4</sub> )   | 29     | 95      |  |  | 8.1  | 7       |
| Acetylene (C <sub>2</sub> H <sub>2</sub> ) | 0      | 2       |  |  | 0.0  | >0      |
| Carbon Monoxyde (CO)                       | 351    | 1100    |  |  | -0.8 | 100     |
| Carbon Dioxyde (CO <sub>2</sub> )          | 3452   | 14000   |  |  | 0.0  | 1000    |
| Oxygen (O <sub>2</sub> )                   | 4002   |         |  |  |      |         |
| Nitrogen (N <sub>2</sub> )                 | 81952  |         |  |  |      |         |

DGA Status = Status 3



# Six Basic Types of Faults

|    |                                                                                |
|----|--------------------------------------------------------------------------------|
| PD | Partial discharges of corona type                                              |
| D1 | Discharges of low energy or partial discharges of sparking type                |
| D2 | Discharges of high energy                                                      |
| T1 | Thermal fault, $t < 300\text{ }^{\circ}\text{C}$                               |
| T2 | Thermal fault, $300\text{ }^{\circ}\text{C} < t < 700\text{ }^{\circ}\text{C}$ |
| T3 | Thermal fault, $t > 700\text{ }^{\circ}\text{C}$                               |

# Additional sub-types of faults

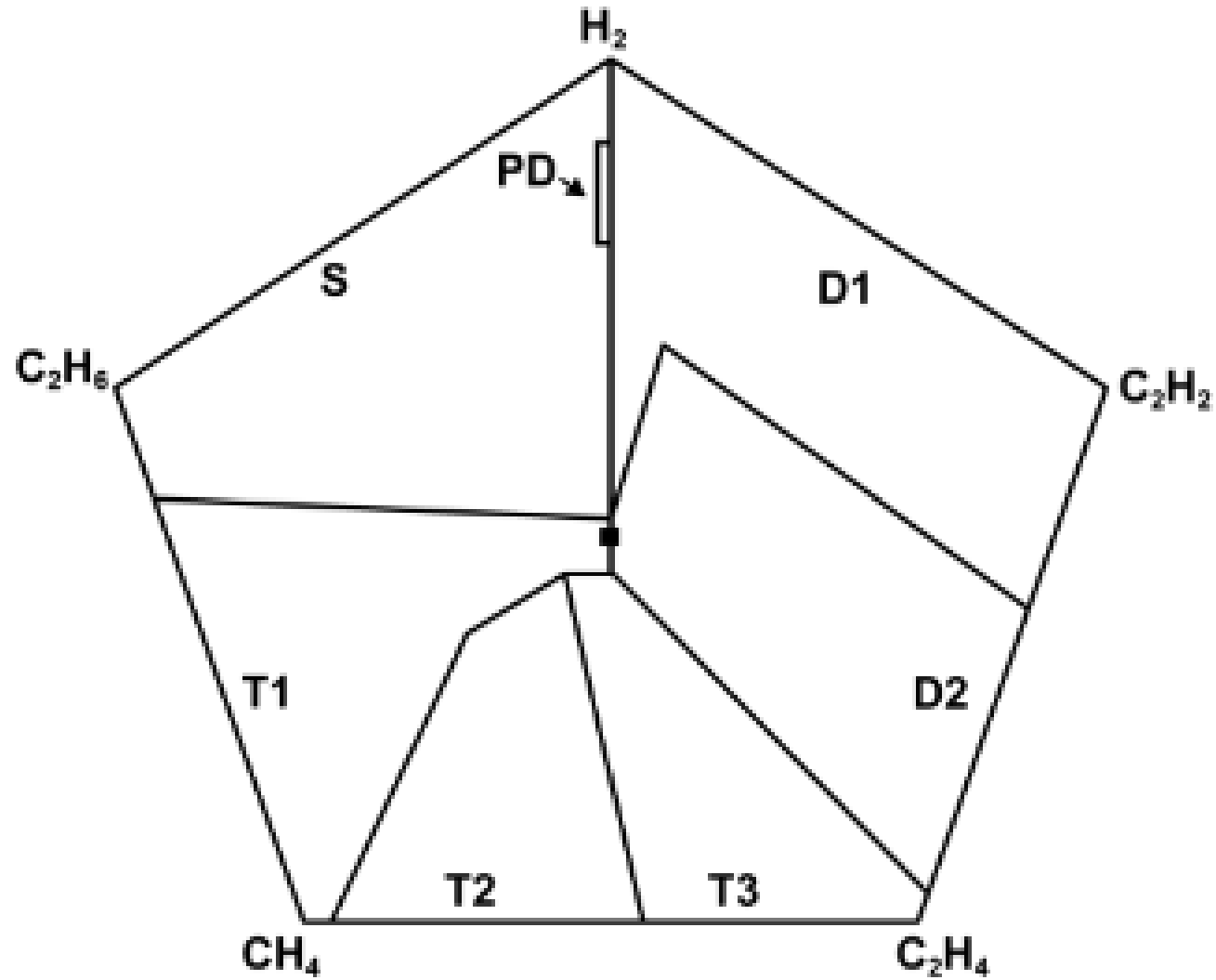
|      |                                                                            |
|------|----------------------------------------------------------------------------|
| S    | Stray gassing at temperatures $< 200\text{ }^{\circ}\text{C}$ ,            |
| O    | Overheating $< 250\text{ }^{\circ}\text{C}$ without carbonization of paper |
| C    | Possible paper carbonization                                               |
| T3-H | Thermal fault T3 in mineral oil only                                       |
| R    | Catalytic reaction                                                         |

- Stray Gassing - terjadi di suhu  $< 200^{\circ}\text{C}$  pada mineral oil, terjadi karena ketidakstabilan dari mineral oil yang dibuat dengan beberapa metode refining baru. Stray gassing juga bisa terjadi karena incompatibility antar material (bisa karena aditif).
- Catalytic reactions - reaksi antara air dan baja pada valve sampling atau air dengan baja pada tangki trafo.

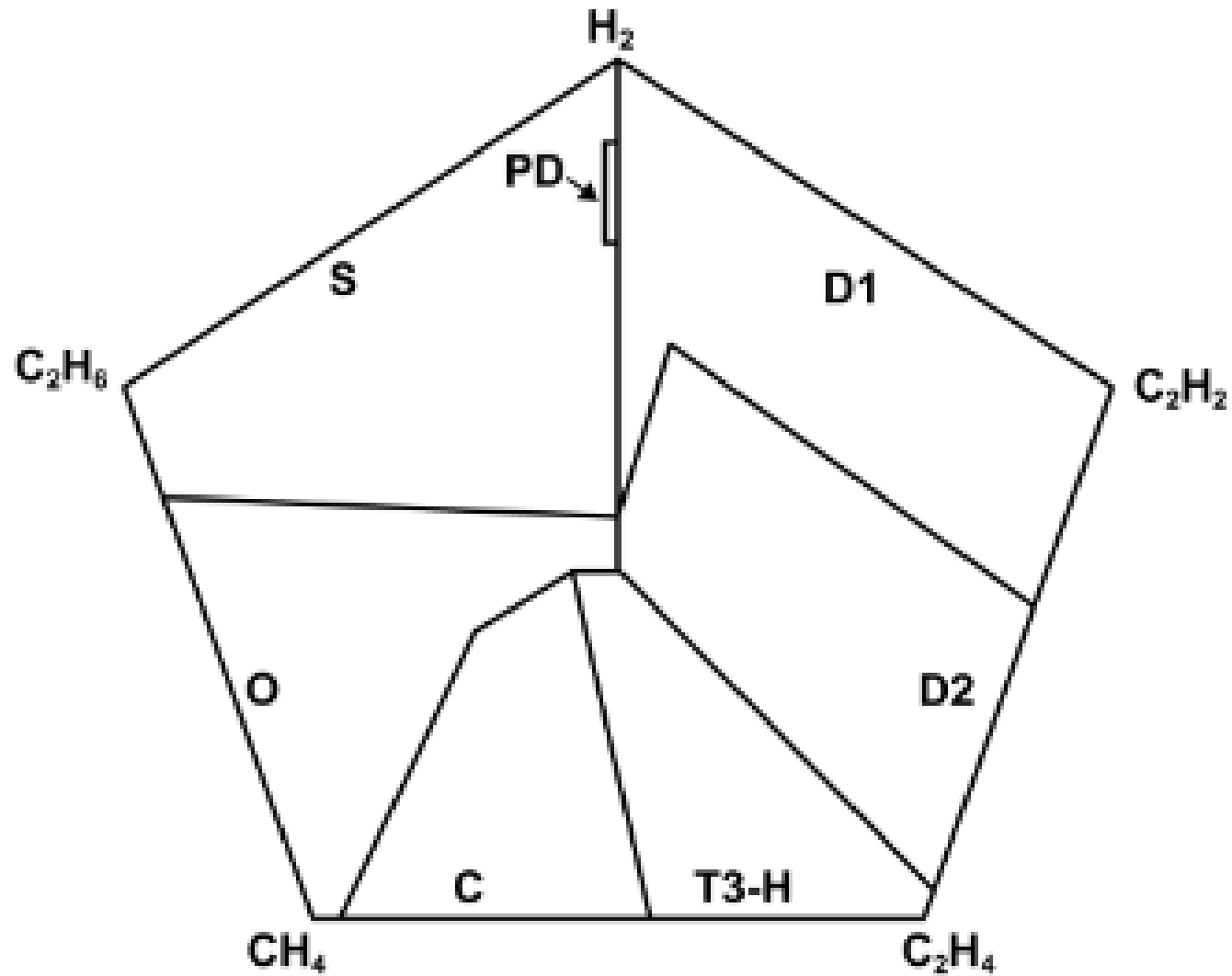
# Metode Interpretasi

- ▶ Six basic types of Faults
  - ▶ Duval Pentagon 1
  - ▶ Duval Triangle 1
- ▶ Additional sub-types of Faults
  - ▶ Duval Pentagon 2
  - ▶ Duval Triangle 4
  - ▶ Duval Triangle 5
- ▶ Mixtures of Faults
  - ▶ Compare Duval Pentagon, Duval Triangle, Keys Gas Ratios & Doenenburg Ratios.

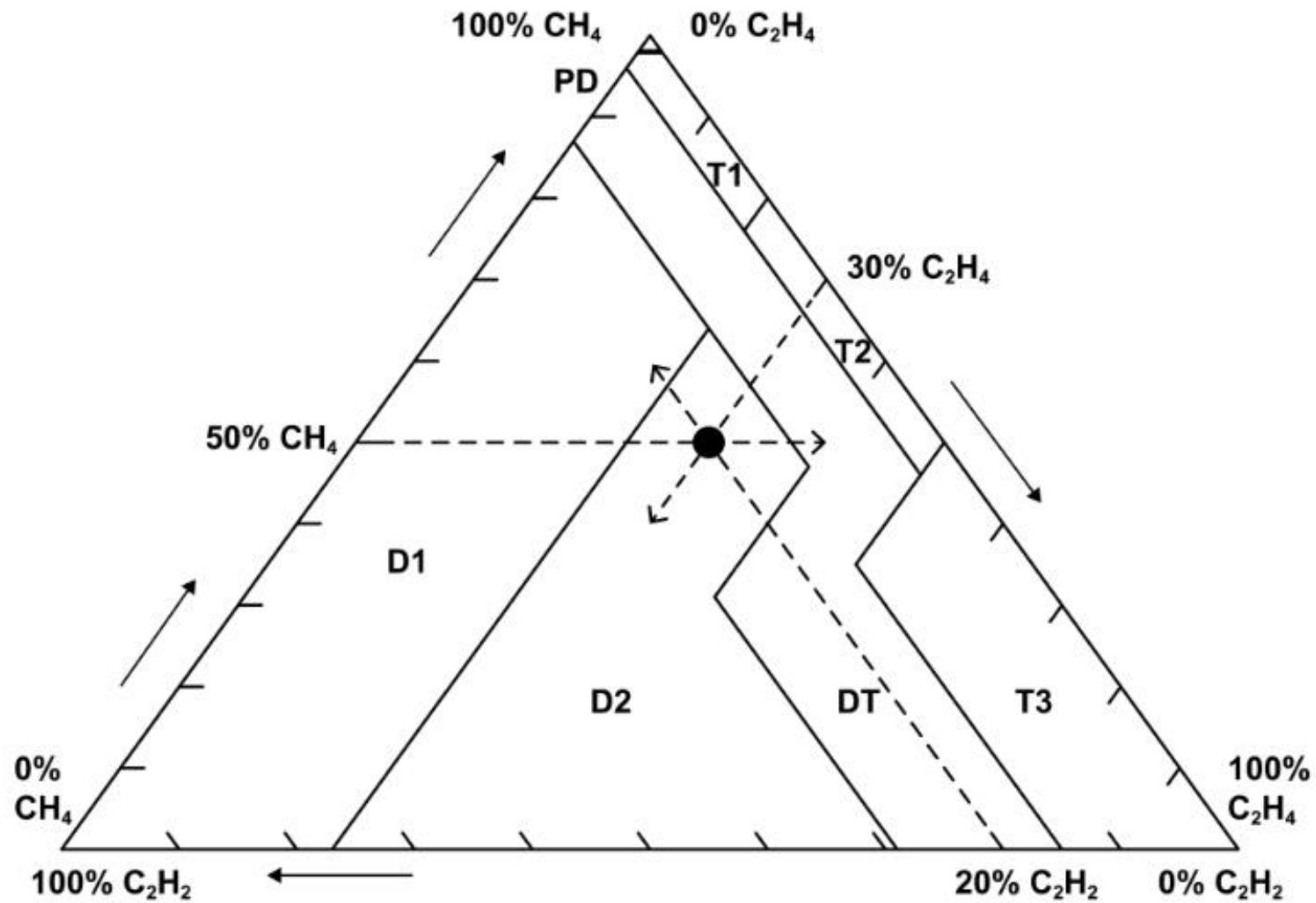
# Duval Pentagon 1



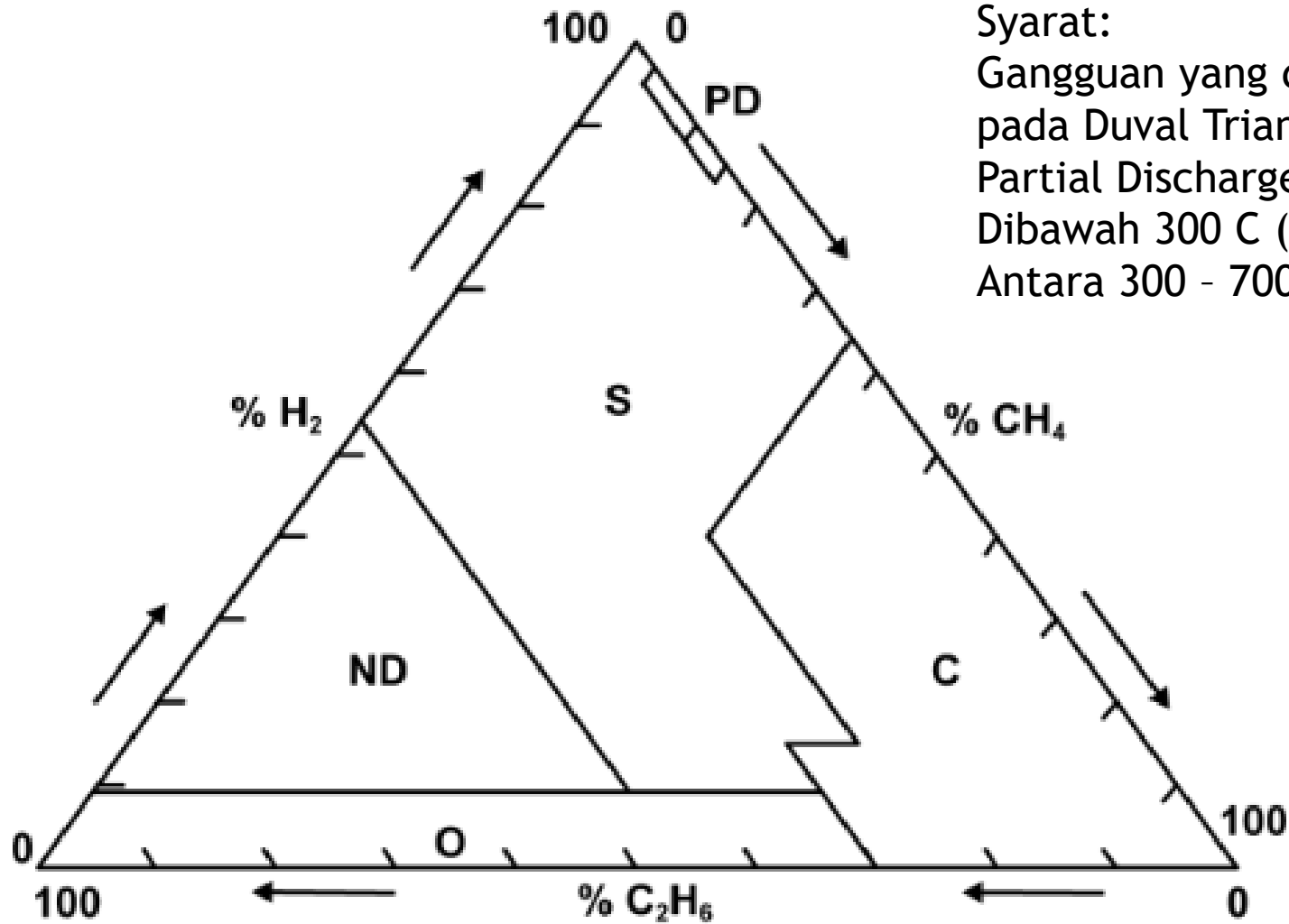
# Duval Pentagon 2



# Duval Triangle 1



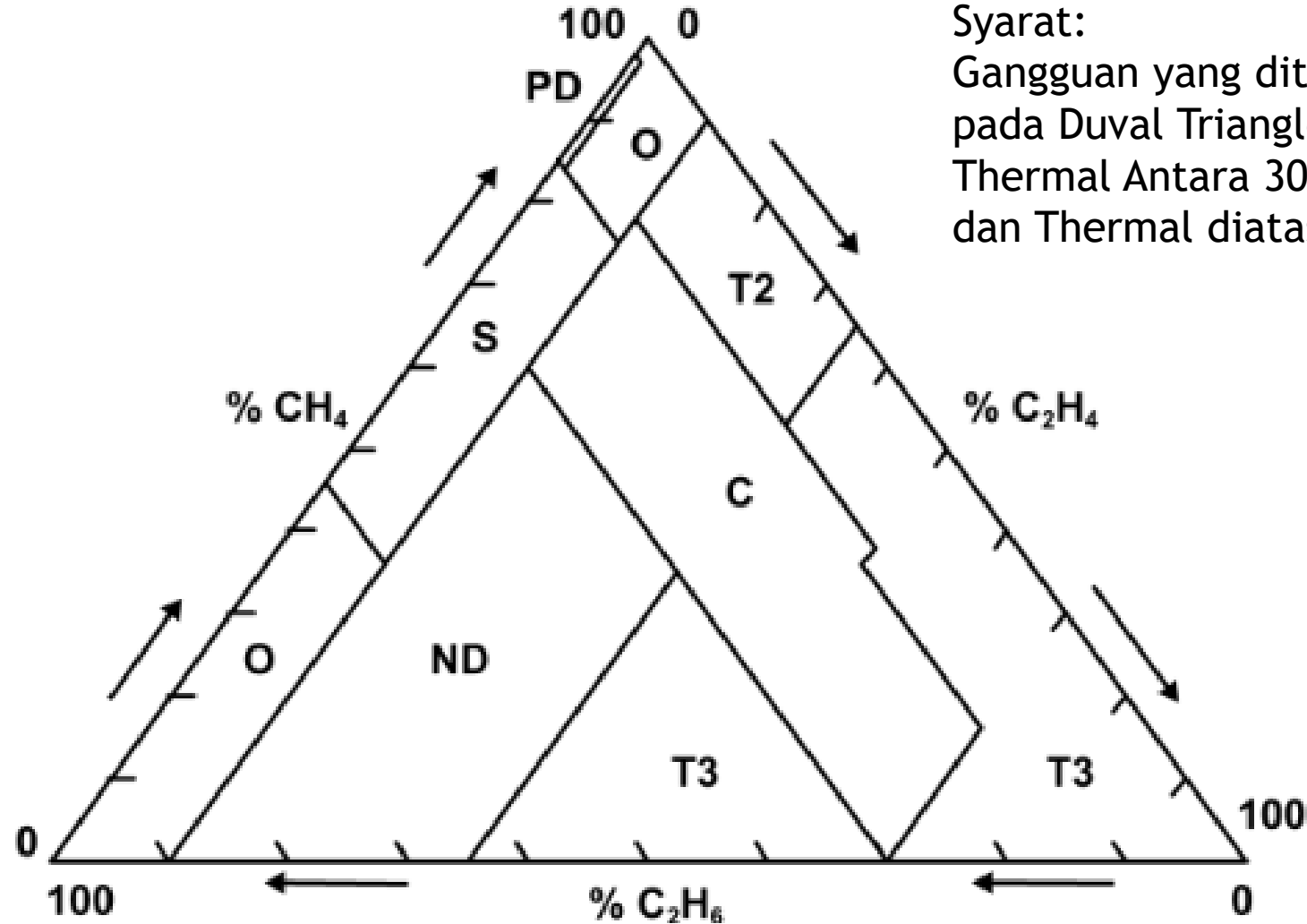
# Duval Triangle 4



Syarat:

Gangguan yang ditunjukkan pada Duval Triangle 1 adalah Partial Discharge (PD), Thermal Dibawah 300 C (T1), dan Thermal Antara 300 - 700 C (T2).

# Duval Triangle 5



Syarat:

Gangguan yang ditunjukkan  
pada Duval Triangle 1 adalah  
Thermal Antara 300 - 700 C (T2)  
dan Thermal diatas 700 C (T3).

# Interpretasi CO dan CO<sub>2</sub>

- ▶ CO > 1000 ppm dan CO<sub>2</sub>/CO < 3, tanpa disertai gas hidrokarbon bukan merupakan indikasi gangguan pada kertas, tetapi lebih disebabkan oleh oksidasi pada minyak dengan kondisi O<sub>2</sub> rendah.
- ▶ CO > 1000 ppm dan CO<sub>2</sub>/CO < 3, disertai gas hidrokarbon dalam jumlah cukup signifikan, mengindikasikan adanya fault pada kertas. Hal ini perlu dikonfirmasi dengan Pentagon 2 dan Triangle 4, 5 dan observasi lain (CO<sub>2</sub> dan Furan)
- ▶ CO<sub>2</sub> > 10000 ppm, CO<sub>2</sub>/CO > 20, dan nilai furan yang tinggi mengindikasikan degradasi kertas pada suhu relative rendah (<140C)
- ▶ Konsentrasi CO dan CO<sub>2</sub> dibawah table 1 menunjukkan normal gassing tanpa fault.
- ▶ Pertumbuhan gas CO dan CO<sub>2</sub> yang rendah tidak selalu menunjukkan tidak adanya gangguan pada kertas. Fault local pada bagian kecil kertas terkadang tidak menghasilkan gas CO dan CO<sub>2</sub> yang terdeteksi. Akan tetapi fault tersebut tetap menghasilkan gas hidrokarbon yang lain dan perlu dilakukan deteksi fault pada kertas dengan Pentagon 2 dan Triangles 4 atau 5.

# DGA FOR OLTC

“Discharges and unwanted arcs cant be detected because the OLTCs work with arcs in oil as their normal operation.”

“Thermal Faults are possible to detect. The arcs during normal switching generate both acetylene and hydrogen but also the three gases indicating thermal faults, methane, ethane and etyhelene.”

# FURAN ANALYSIS

## IEC 61198

Pengujian untuk mengetahui kondisi isolator  
kertas

# PERAN ISOLATOR KERTAS

Isolator kertas pada trafo merupakan isolator utama pada trafo, usia dari trafo sama dengan usia dari isolator kertas tersebut. Kenaikan kandungan air pada kertas trafo sebesar 2 kali akan menurunkan usia isolator kertas menjadi setengahnya.

Seiring usia dari trafo, isolator kertas akan mengalami degradasi, kandungan air pada kertas bisa mempercepat proses degradasi tersebut.

# FURAN ANALYSIS

Degradasi kertas pada trafo akan terlarut pada minyak trafo.

Furan analysis digunakan untuk menghitung seberapa banyak kandungan degradasi kertas yang terlarut pada minyak trafo.

Usia kertas =  $100\% - \text{kertas terlarut}$



**Table – Furans, DP, Percent of Life Used of Paper Insulation**

| 55 °C Rise<br>Transformer<br>2FAL<br>(ppb) | 65 °C Rise<br>Trans-former<br>Total Furans<br>(ppb) | Estimated<br>Degree of<br>Polymerization<br>(DP) | Estimated<br>Percentage<br>of Remaining<br>Life | Interpretation                                                           |
|--------------------------------------------|-----------------------------------------------------|--------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------|
| 58                                         | 51                                                  | 800                                              | 100                                             | <b>Normal Aging<br/>Rate</b>                                             |
| 130                                        | 100                                                 | 700                                              | 90                                              |                                                                          |
| 292                                        | 195                                                 | 600                                              | 79                                              |                                                                          |
| 654                                        | 381                                                 | 500                                              | 66                                              | <b>Accelerated<br/>Aging Rate</b>                                        |
| 1,464                                      | 745                                                 | 400                                              | 50                                              |                                                                          |
| 1,720                                      | 852                                                 | 380                                              | 46                                              |                                                                          |
| 2,021                                      | 974                                                 | 360                                              | 42                                              |                                                                          |
| 2,374                                      | 1,113                                               | 340                                              | 38                                              | Excessive Aging<br>Danger Zone                                           |
| 2,789                                      | 1,273                                               | 320                                              | 33                                              |                                                                          |
| 3,277                                      | 1,455                                               | 300                                              | 29                                              |                                                                          |
| 3,851                                      | 1,664                                               | 280                                              | 24                                              | High Risk of<br>Failure                                                  |
| 4,524                                      | 1,902                                               | 260                                              | 19                                              |                                                                          |
| 5,315                                      | 2,175                                               | 240                                              | 13                                              | End of Expected<br>Life of Paper<br>Insulation and of<br>the Transformer |
| 6,245                                      | 2,487                                               | 220                                              | 7                                               |                                                                          |
| 7,337                                      | 2,843                                               | 200                                              | 0                                               |                                                                          |

# OIL QUALITY ANALYSIS

Pengujian untuk mengetahui kualitas minyak  
transformator

# OIL QUALITY ANALYSIS

- ▶ Breakdown Voltage (Tegangan Tembus)
- ▶ Water Content (Kandungan Air)
- ▶ Total Acid Number (Keasaman)
- ▶ Color & Appearance (Warna dan Tampilan)
- ▶ Interfacial Tension (Tegangan Antarmuka)
- ▶ Dielectric Dissipation Factor & Resistivity
- ▶ Metal Content
- ▶ Corrosive Sulphur
- ▶ Flash Point (Titik Nyala)
- ▶ OQIN (Oil Quality Index Number)

# BREAKDOWN VOLTAGE / TEGANGAN TEMBUS (IEC 60156)

Pengujian untuk mengetahui kemampuan dielektrik dari minyak trafo, berapa besar tegangan yang bisa ditahan oleh minyak trafo.

BDV sendiri dipengaruhi oleh kandungan air dan kontaminan pada minyak.



# BATASAN NILAI BREAKDOWN VOLTAGE (IEC 60296 & IEC 60422)

| NEW OIL            |                    |
|--------------------|--------------------|
| Sebelum Purifikasi | Setelah Purifikasi |
| Min. 30 kV         | Min 70 kV          |

| NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU |               |            |
|--------------------------------------------|---------------|------------|
| < 72.5 kV                                  | 72.5 - 170 kV | > 170 kV   |
| Min. 55 kV                                 | Min. 60 kV    | Min. 60 kV |

| USED OIL |            |               |            |
|----------|------------|---------------|------------|
|          | < 72.5 kV  | 72.5 - 170 kV | > 170 kV   |
| Good     | > 40 kV    | > 50 kV       | > 60 kV    |
| Fair     | 30 - 40 kV | 40 - 50 kV    | 50 - 60 kV |
| Poor     | < 30 kV    | < 40 kV       | < 50 kV    |

# WATER CONTENT / KANDUNGAN AIR (IEC 60814)

Air merupakan hal yang paling dihindari ada di dalam trafo. Kandungan air pada minyak trafo secara langsung akan mempengaruhi nilai BDV dan dalam jangka panjang akan memicu reaksi kimia pada minyak sehingga menurunkan kualitas minyak untuk parameter yang lain.



# BATASAN NILAI WATER CONTENT (IEC 60296 & IEC 60422)

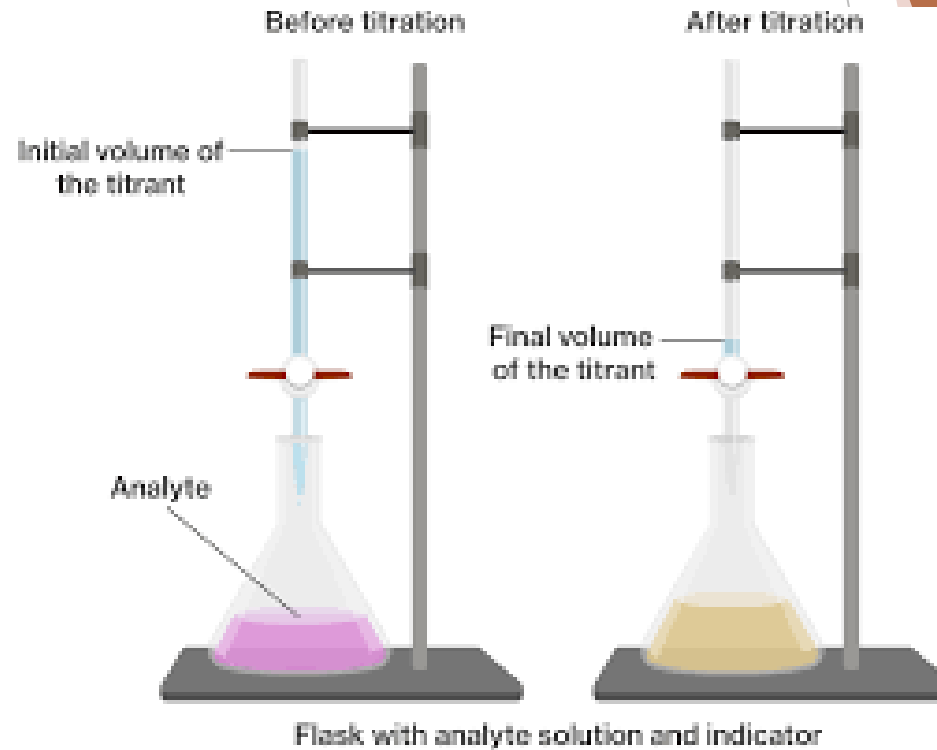
| NEW OIL                       |                              |
|-------------------------------|------------------------------|
| Pengiriman dalam jumlah besar | Pengiriman dengan drum / IBC |
| Max. 30 ppm                   | Max. 40 ppm                  |

| NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU |               |             |
|--------------------------------------------|---------------|-------------|
| < 72.5 kV                                  | 72.5 - 170 kV | > 170 kV    |
| Max. 20 ppm                                | Max. 10 ppm   | Max. 10 ppm |

| USED OIL |             |               |             |
|----------|-------------|---------------|-------------|
|          | < 72.5 kV   | 72.5 - 170 kV | > 170 kV    |
| Good     | < 30 ppm    | < 20 ppm      | < 15 ppm    |
| Fair     | 30 - 40 ppm | 20 - 30 ppm   | 15 - 20 ppm |
| Poor     | > 40 ppm    | > 30 ppm      | > 20 ppm    |

# TOTAL ACID NUMBER / KEASAMAN (IEC 62021-2)

Keasaman adalah salah satu produk dari reaksi kimia (oksidasi) pada minyak trafo. Oksidasi pada minyak trafo dipicu oleh adanya air, oksigen dan panas yang terjadi di dalam trafo.



# BATASAN NILAI TOTAL ACIDITY (IEC 60296 & IEC 60422)

## NEW OIL

Max. 0.01 mg KOH/g

## NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU

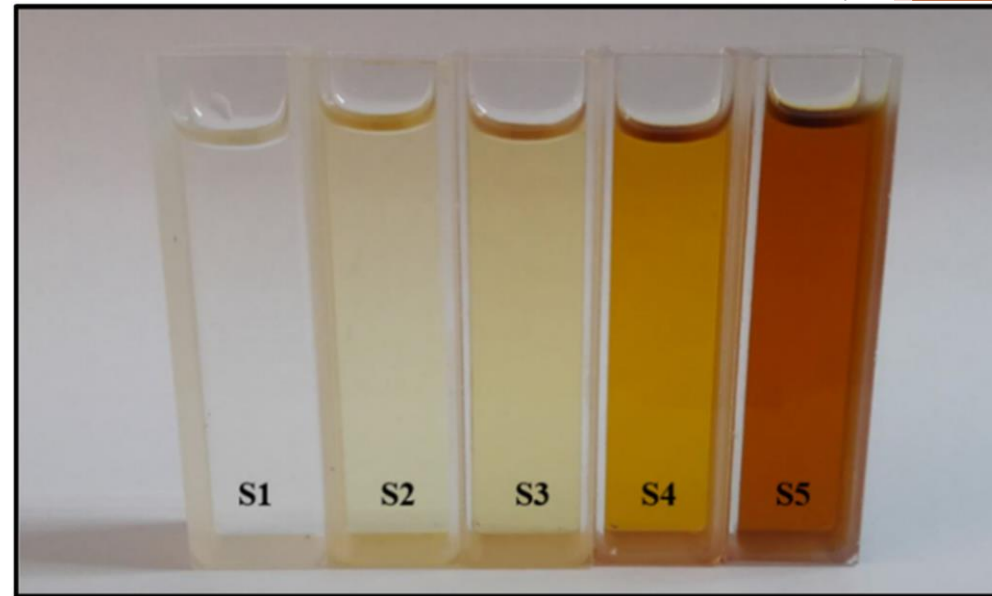
Max. 0.03 mg KOH/g

## USED OIL

|      | < 72.5 kV   | 72.5 - 170 kV | > 170 kV    |
|------|-------------|---------------|-------------|
| Good | < 0.15      | < 0.10        | < 0.10      |
| Fair | 0.15 - 0.30 | 0.10 - 0.20   | 0.10 - 0.15 |
| Poor | > 0.30      | > 0.20        | > 0.15      |

## COLOR & APPEARANCE / WARNA & TAMPILAN (ASTM D1500)

Warna minyak trafo akan menjadi gelap seiring terjadinya reaksi kimia. Hasil oksidasi pada minyak trafo akan menghasilkan senyawa-senyawa kimia lain yang menjadikan warna minyak menjadi gelap.



# BATASAN NILAI TOTAL ACIDITY (IEC 60296 & IEC 60422)

## NEW OIL

Clear, free from sediment and suspended matter

## NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU

Clear, free from sediment and suspended matter

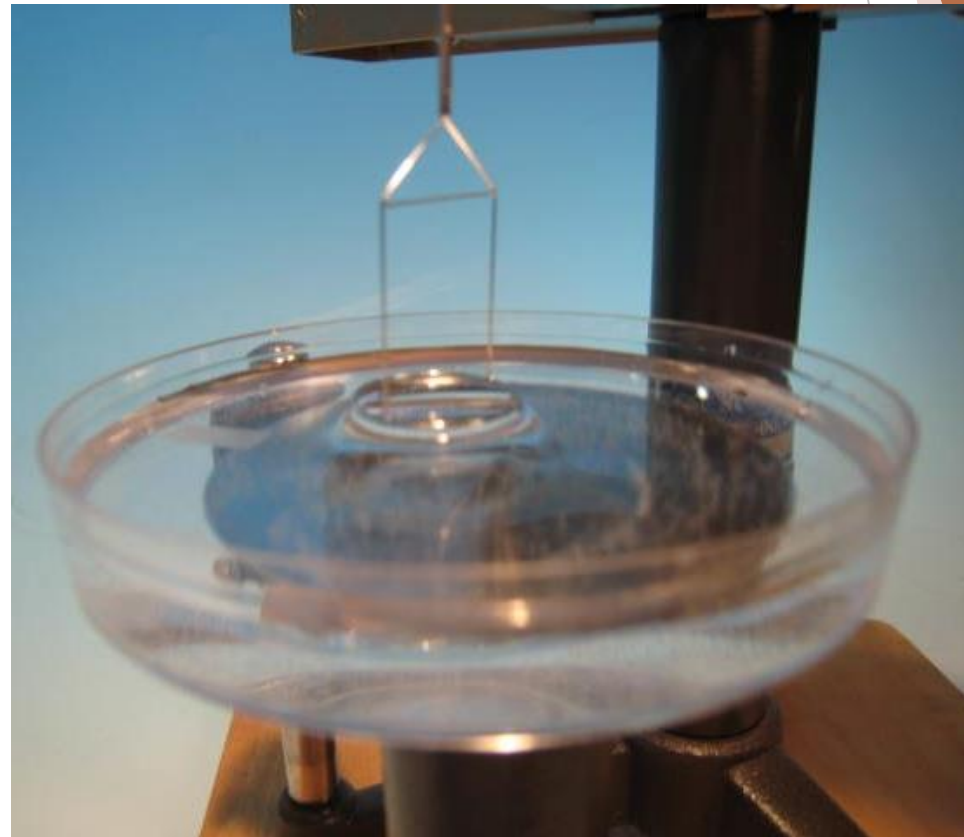
## USED OIL

|      | < 72.5 kV                               | 72.5 - 170 kV | > 170 kV |
|------|-----------------------------------------|---------------|----------|
| Good | Clear and without visible contamination |               |          |
| Fair | -                                       |               |          |
| Poor | Dark and/or turbid                      |               |          |

# INTERFACIAL TENSION / TEGANGAN ANTARMUKA (ASTM D971)

IFT merupakan kemampuan minyak untuk menolak air, sehingga ketika IFT minyak tinggi, maka air akan susah tercampur dengan minyak.

Kualitas IFT sendiri akan turun dengan adanya oksidasi pada minyak serta kandungan air pada minyak juga dapat menurunkan kualitas IFT.



# BATASAN NILAI TOTAL ACIDITY (IEC 60296 & IEC 60422)

## NEW OIL

No general requirement

## NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU

Min. 35 mN/m

## USED OIL

|      | Inhibit Oil | Un-Inhibit Oil |
|------|-------------|----------------|
| Good | > 28        | > 25           |
| Fair | 22 - 28     | 20 - 25        |
| Poor | < 22        | < 20           |

# DIELECTRIC DISSIPATION FACTOR & RESISTIVITY at 90° C (IEC 60247)

Dielectric Dissipation Factor & Resistivity merupakan pengujian untuk mengetahui tingkat disipasi dan resistivitas minyak trafo.

Parameter ini sensitive terhadap kontaminan polar yang bisa larut atau penuaan dari minyak.

Kualitas parameter ini dapat mempengaruhi dielectric losses dan/atau kemampuan isolator dari peralatan listrik.

# BATASAN NILAI DIELECTRIC DISSIPATION FACTOR (IEC 60296 & IEC 60422)

## NEW OIL

Max. 0,005

## NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU

| < 72.5 kV  | 72.5 - 170 kV | > 170 kV   |
|------------|---------------|------------|
| Max. 0.015 | Max. 0.015    | Max. 0.010 |

## USED OIL

|      | < 72.5 kV   | 72.5 - 170 kV | > 170 kV    |
|------|-------------|---------------|-------------|
| Good | < 0.10      | < 0.10        | < 0.10      |
| Fair | 0.10 - 0.50 | 0.10 - 0.50   | 0.10 - 0.20 |
| Poor | > 0.50      | > 0.50        | > 0.20      |

# BATASAN NILAI RESISTIVITY (IEC 60296 & IEC 60422)

## NEW OIL

Tidak dijelaskan

## NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU

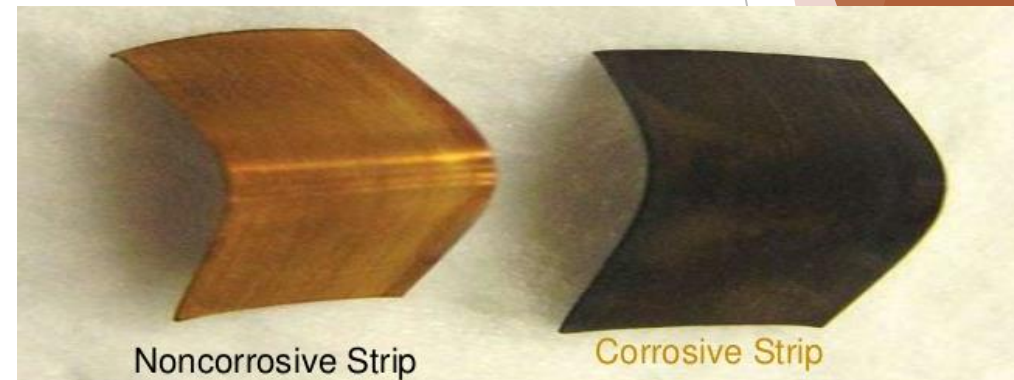
Min. 60 GΩm

## USED OIL

|      | < 72.5 kV | 72.5 - 170 kV | > 170 kV |
|------|-----------|---------------|----------|
| Good | > 3       | > 3           | > 10     |
| Fair | 0.2 - 3   | 0.2 - 3       | 3 - 10   |
| Poor | < 0.2     | < 0.2         | < 3      |

# CORROSIVE SULPHUR (ASTM D1275-B)

Minyak trafo merupakan salah satu hasil dari pengolahan minyak bumi, dimana minyak bumi sendiri memiliki kandungan sulfur. Dengan adanya kandungan sulfur pada minyak trafo, kemampuan korosif dari minyak trafo harus dikendalikan. Efek korosif pada minyak trafo akan merusak bagian-bagian pada trafo terutama konduktor-konduktor yang ada di dalam trafo.



# BATASAN NILAI CORROSIVE SULPHUR (IEC 60296 & IEC 60422)

## NEW OIL

Non Corrosive

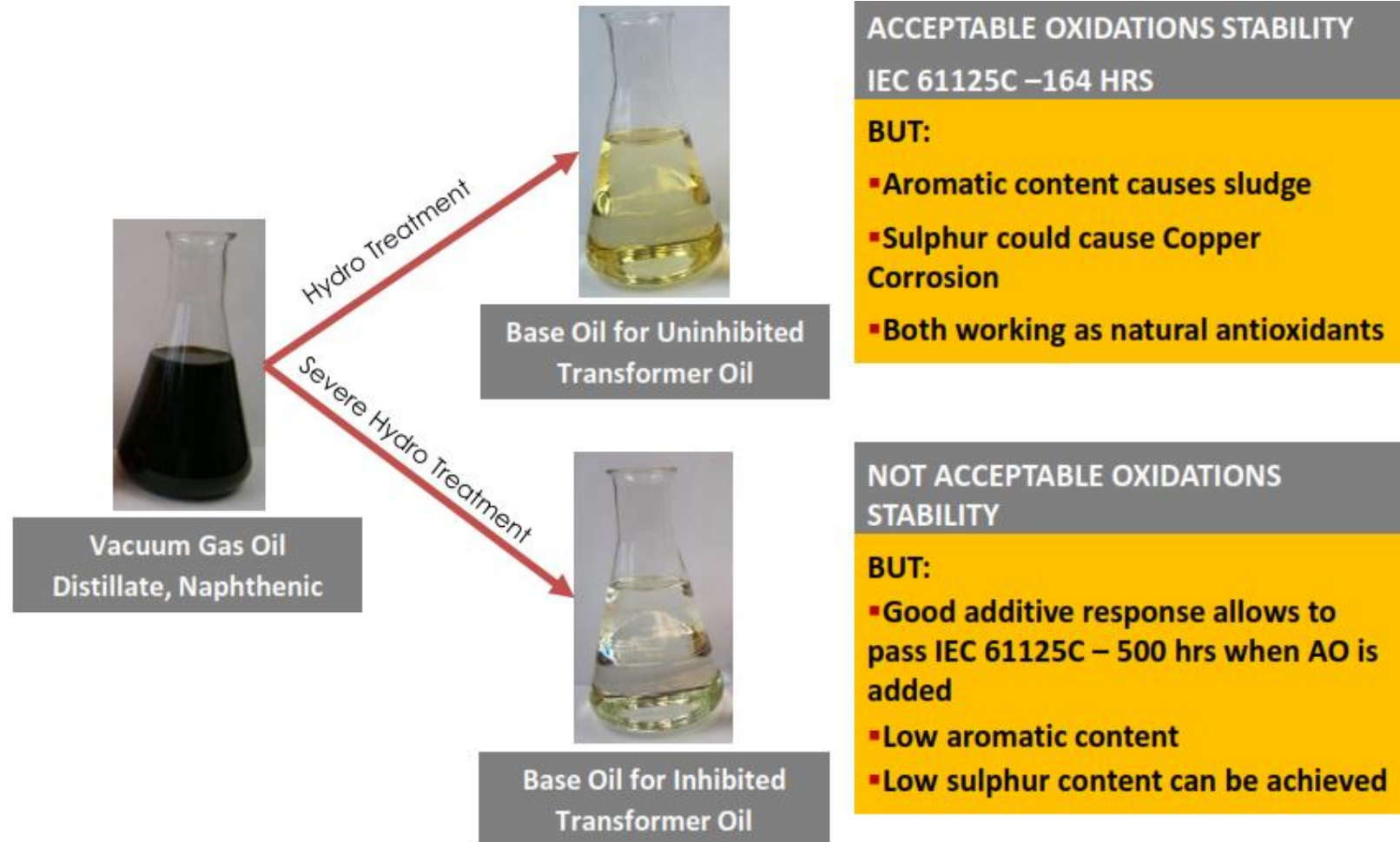
## NEW OIL SETELAH DIMASUKAN DALAM TRAFO BARU

Non Corrosive

## USED OIL

|      | < 72.5 kV     | 72.5 - 170 kV | > 170 kV |
|------|---------------|---------------|----------|
| Good | Non Corrosive |               |          |
| Fair | -             |               |          |
| Poor | Corrosive     |               |          |

# INHIBITOR CONTENT (IEC 61125)



# Jenis Minyak Dari Kandung Inhibitor

- ▶ Un-inhibited Oil : Tidak mengandung zat aditif
- ▶ Trace Inhibit : <0.08%
- ▶ Inhibit Oil : 0.1% - 0.4%

# FILTRASI / PURIFIKASI

Filtrasi atau Purifikasi dilakukan untuk mengurangi kontaminasi fisik, kandungan air dan gas. Dalam proses filtrasi dilakukan 3 tahapan, yaitu:

- ▶ Filtrasi (penyaringan minyak) untuk menghilangkan kontaminasi fisik.
- ▶ De-humidifikasi untuk menghilangkan kandungan air
- ▶ De-gasifikasi untuk menghilangkan kandungan gas terlarut

Proses de-humidifikasi dan de-gasifikasi menggunakan metode vakum sehingga tidak akan terkontaminasi dari luar.



# TERIMA KASIH

FIKRI KHOIRUR RIZAL ABDUL QOHAR  
PT TRANSFORMATOR SERVIS INDONESIA